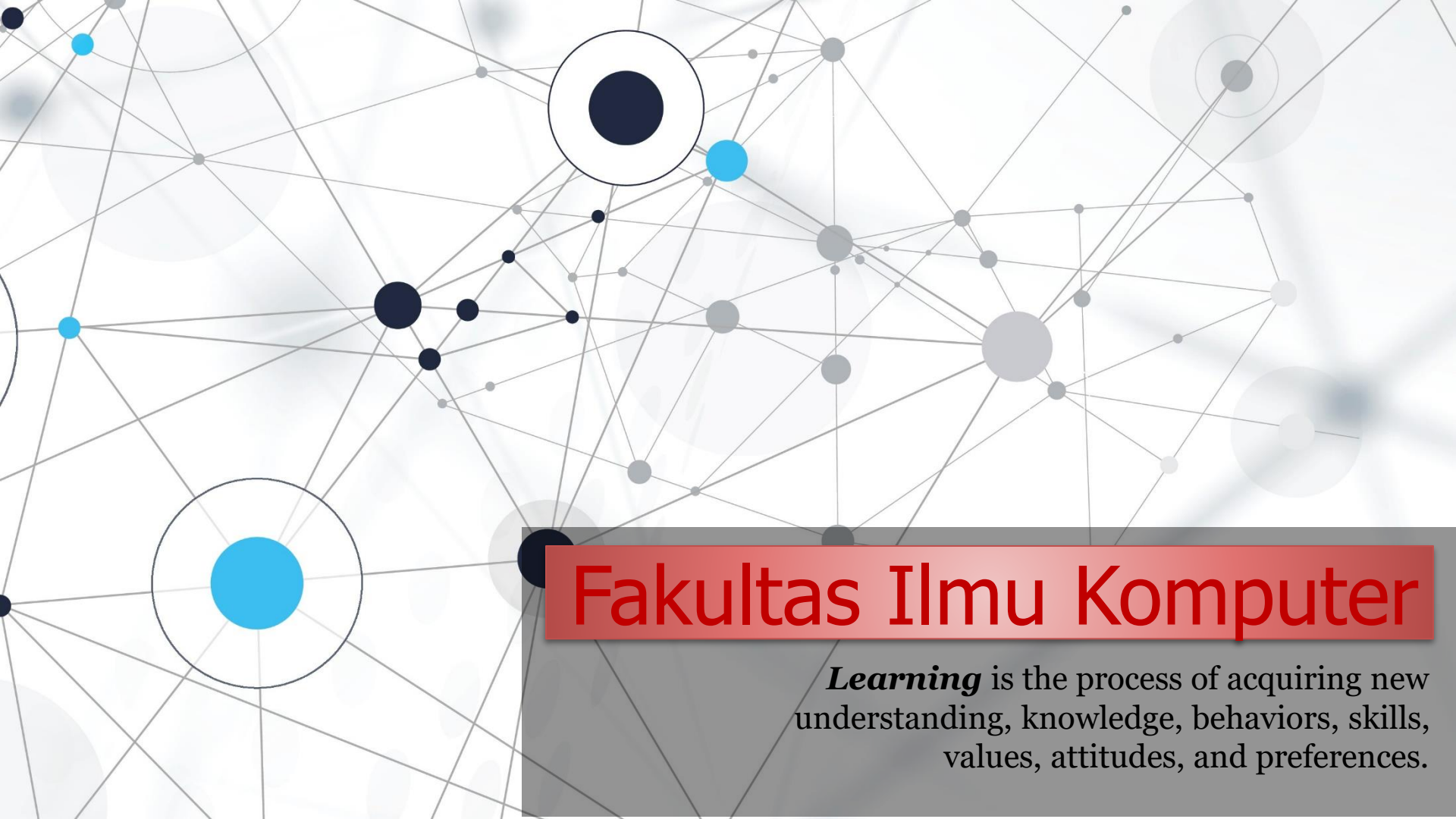


An abstract network diagram with various sized nodes (black, blue, grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

Source Code Management (SCM) DevOps Version Control System (VCS)

Menggunakan Git & GitHub (2023)



**Welcome to *the* DevOps
Engineering course**

Di class practical sebelumnya kalian sudah melakukan cara manual deploy/edit code di sisi server. Menurut kalian apakah mudah atau sulit?

Kalian bisa bayangkan kalian diminta untuk berkontribusi di dua tim (dev dan ops) tersebut. Pasti ada banyak tugas yang harus kalian kerjakan.

remote/central repo



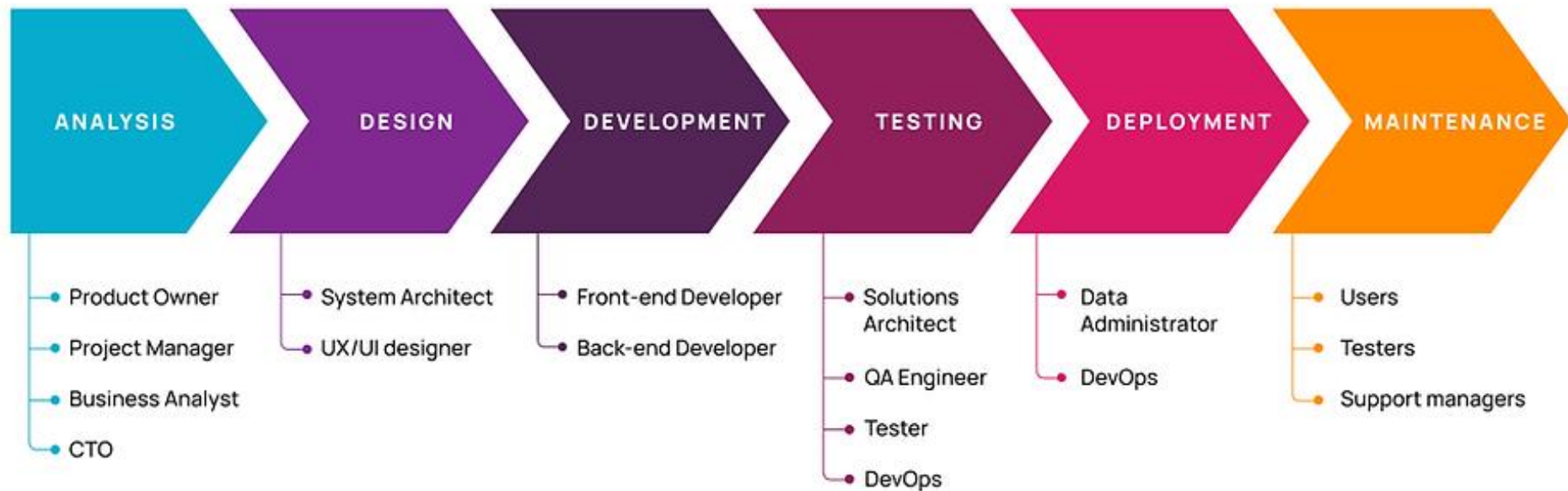
local repositories

Chapter 4: Everything is Code	39
The need for source code control	40
The history of source code management	40
Roles and code	41
Which source code management system?	42
A word about source code management system migrations	43
Choosing a branching strategy	43
Branching problem areas	45
Artifact version naming	46
Choosing a client	47
Setting up a basic Git server	48
Shared authentication	50
Hosted Git servers	50
Large binary files	51
Trying out different Git server implementations	52
Docker intermission	52
Gerrit	53
Installing the git-review package	54
The value of history revisionism	55



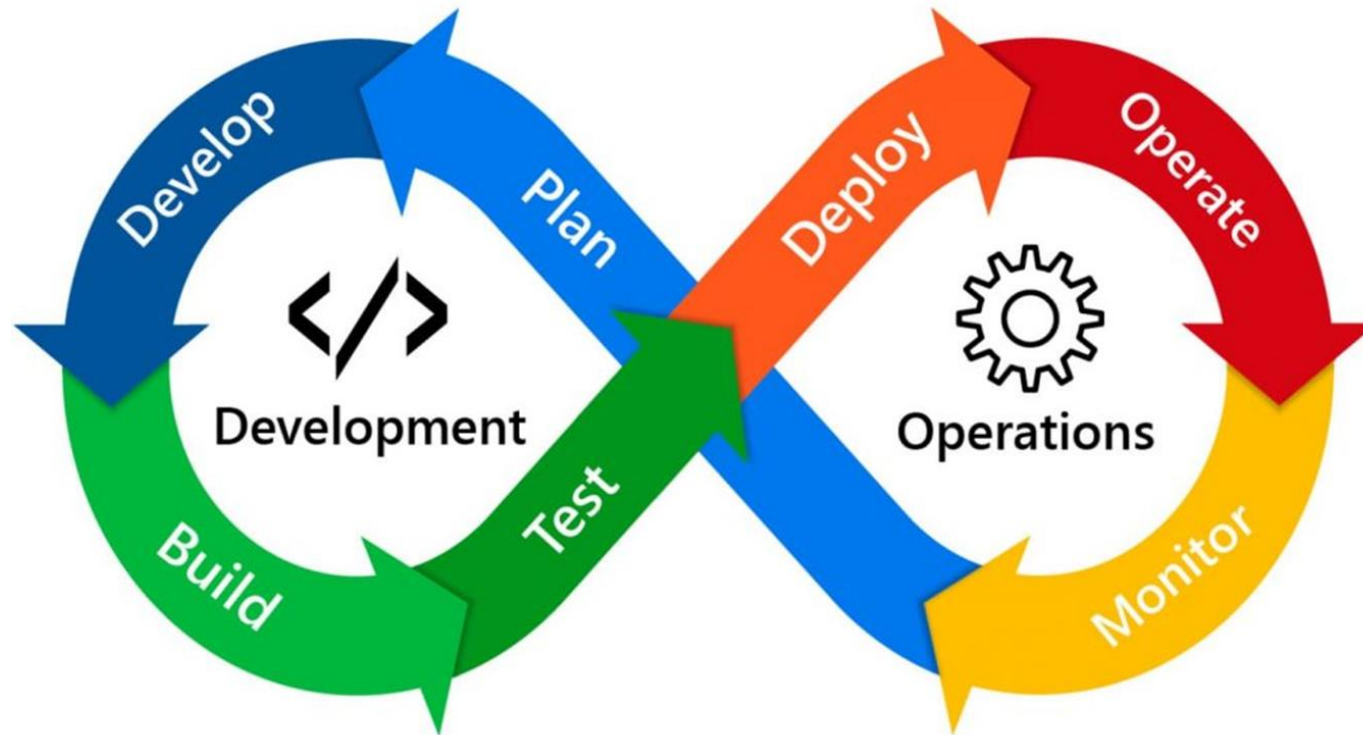
Teamwork is essential in software development, as software development projects are complex processes involving many elements that must work together efficiently.

System Development Life Cycle (SDLC)



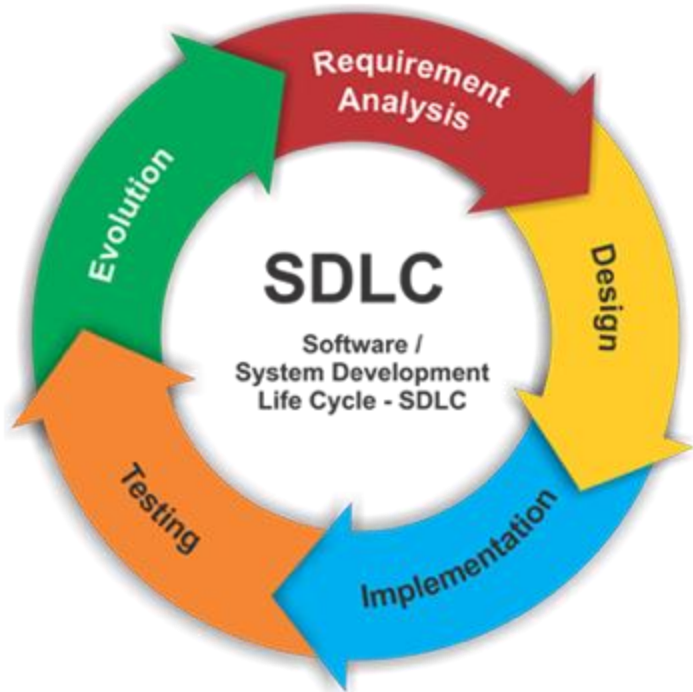
The Software Development Life Cycle (SDLC) is a structured process that enables the production of **high-quality, low-cost software**, **in the shortest possible production time**. The goal of the SDLC is **to produce superior software** that meets and exceeds all customer expectations and demands/requests.

DevOps Methodology



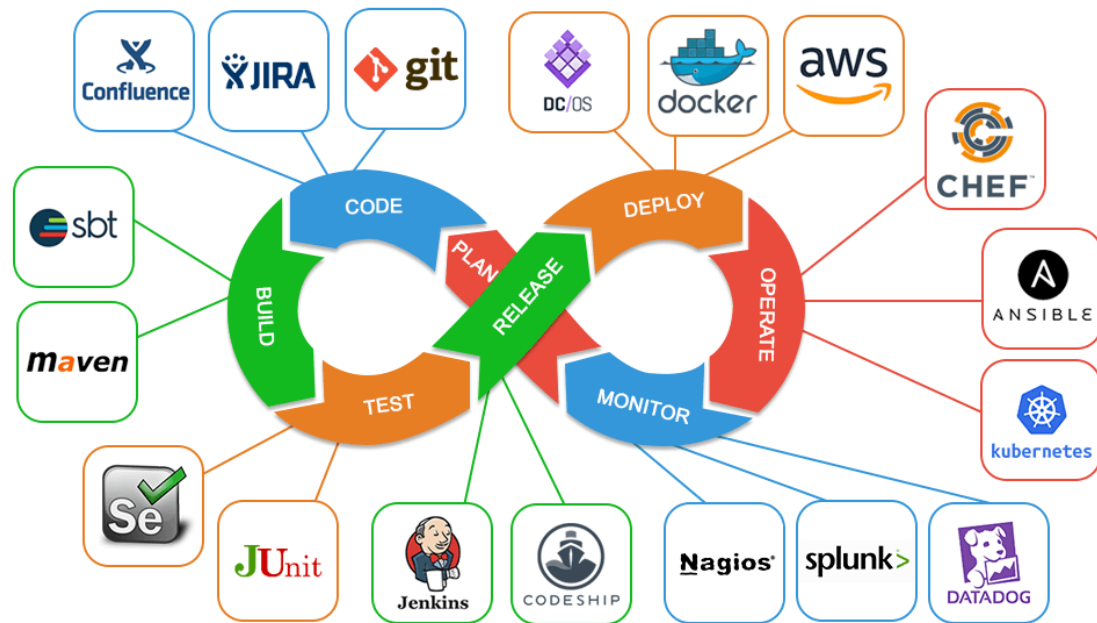
CI/CD stands for Continuous Integration and Continuous Delivery. CI/CD is an effort to **change the software development process from manual to automated**. CI/CD is a software development method that facilitates, accelerates, and improves the quality of the final software released.

(Before) Automation Servers



- 1) Dalam pengembangan software (*perangkat lunak*), koding hanyalah tahapan awal saja.
- 2) Setelah koding, masih banyak hal yang biasanya dilakukan oleh *software developer* (*perangkat lunak*) sebelum aplikasi yang dibuat sampai ke *production server*.
- 3) Sebelum adanya Automation Server, biasanya semuanya dilakukan **secara manual**, namun resikonya adalah sering terjadi **kesalahan dan tidak konsisten**.

Contoh menjalankan ***unit test***, download dependency aplikasi, setup database, membuat distribusi aplikasi, dan lain-lain. Hal-hal diatas sangat kompleks jika harus dilakukan secara manual terus menerus.



- 1) **Automation Server** adalah tool/aplikasi yang bisa kita gunakan untuk **melakukan proses otomatisasi perintah** yang kita instruksikan.
- 2) Ada banyak sekali keuntungan yang kita dapatkan jika menggunakan Automation Server, dibanding manual dalam tahapan pengembangan software (perangkat lunak).
- 3) (a) Automation Server **mempercepat proses**, karena kita tidak perlu menunggu di Laptop kita.
(b) Automation Server bisa **berjalan konsisten**, di Laptop programmer mungkin perangkat lunak bisa berbeda-beda.
- 4) Automation Server bisa **berjalan secara otomatis** (hampir sama seperti *scheduler*), bisa memberitahu kita ketika terjadi kesalahan, dan lain-lain.

Version Control (pengendalian versi) adalah komponen penting dalam CI/CD (Continuous Integration/Continuous Deployment) dalam proses DevOps. Ini sangat diperlukan karena membawa sejumlah manfaat yang krusial untuk **manajemen kode**, **pengembangan perangkat lunak yang efisien**, dan **otomatisasi proses pengiriman**.

Manajemen Konflik

Ketika beberapa pengembang bekerja dalam tim pada proyek yang sama, konflik dalam perubahan kode adalah hal yang umum. Version Control menyediakan mekanisme untuk menangani konflik ini dengan aman dan meresolusi perubahan kode yang bertentangan.





1 – SCM *vs* VCS

Source Code Management (SCM)

- ❖ Source code management (SCM) is used to track modifications to a source code repository. SCM tracks a running history of changes to a code base and helps resolve conflicts when merging updates from multiple contributors.
- ❖ **SCM is also synonymous with Version control.** SCM is a critical tool to alleviate the organizational strain of growing development costs.
- ❖ Source Code Management (SCM) dalam konteks DevOps merujuk pada praktik-praktik dan alat-alat yang digunakan untuk mengelola **source code** dari aplikasi perangkat lunak selama seluruh siklus hidup pengembangan perangkat lunak.
- ❖ **SCM merupakan salah satu komponen penting dalam proses DevOps karena memungkinkan tim pengembangan untuk bekerja sama secara efisien, melacak perubahan kode, dan mengotomatisasi aliran kerja pengembangan perangkat lunak.**

Version Control System (VCS)

- ❖ Version control system (VCS) adalah sebuah sistem yang **mencatat setiap perubahan terhadap sebuah berkas atau kumpulan berkas (kode)** sehingga pada suatu saat developer (*pengguna*) dapat kembali kepada ke salah satu versi dari berkas (*kode*) yang diinginkan.
- ❖ VCS sangat mendukung pengembangan *software* yang kolaboratif. Setiap anggota tim development dapat menulis kode programnya masing-masing yang kemudian akan digabungkan menggunakan VCS ini.
- ❖ VCS memperbolehkan setiap *developer* untuk
 - a) Mengembalikan suatu berkas kode ke keadaan sebelumnya,
 - b) Mengembalikan keseluruhan proyek ke keadaan sebelumnya,
 - c) Membandingkan perubahan-perubahan berkas kode dari setiap waktu,
 - d) Melihat siapa orang yang terakhir mengubah kode yang mungkin menimbulkan konflik, kapan isu muncul, siapa yang memunculkan sebuah isu,
 - e) *dll.*

7 Version Control Systems (VCS)

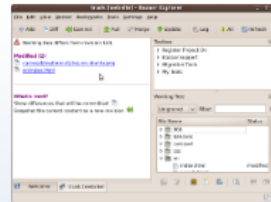
Terdapat banyak Version Control System yang telah dikembangkan para developer. Ada yang berbayar dan ada juga yang free/gratis.

[1] Bazaar merupakan bagian dari GNU Project, Bazaar adalah free software yang disponsori oleh Canonical. Salah satu layanan yang menggunakan Bazaar adalah Launchpad, sebuah tempat dimana aplikasi Ubuntu dikembangkan dan dipantau oleh komunitas. Bazaar dapat digunakan di Windows, Ubuntu, Debian, Red Hat, SUSE, OS X, FreeBSD, Solaris, Gentoo, dan lainnya.



Bazaar

Stable version: 2.7.0
Test version: trunk



Take the tour on [GNOME](#), [KDE](#), [Windows](#) or [OS X](#).

What is Bazaar?

Bazaar is a version control system that helps you track project history over time and to collaborate easily with others. Whether you're a single developer, a co-located team or a community of developers scattered across the world, Bazaar scales and adapts to meet your needs. Part of the **GNU Project**, Bazaar is free software sponsored by **Canonical**. For a closer look, see [ten reasons to switch to Bazaar](#).

🌱 Downloads

Bazaar runs on [Windows](#), [Ubuntu](#), [Debian](#), [SUSE](#), [Fedora](#), [OS X](#), [FreeBSD](#), [Solaris](#), [Gentoo](#), and more.

✚ Extras

Get your team working better together with project hosting, add-ons and open APIs for custom integration needs.

🔍 Support

Our active community can be found on IRC and the project mailing list.

[Get Bazaar](#)

[Get Going](#)

[Get Help](#)

Community News

- 30-Sep-2017: [bzt.mk](#) - A Makefile for generating `.bzrignore`
- 09-Apr-2016: [QBzr 0.23.2](#) released
- 15-Feb-2016: [2.7.0](#) released
- 31-Jan-2014: [bzt-eclipse 1.4](#) released

Core News

- 15-Feb-2016: [2.7.0](#) released
- 04-Aug-2013: [2.6.0](#) released

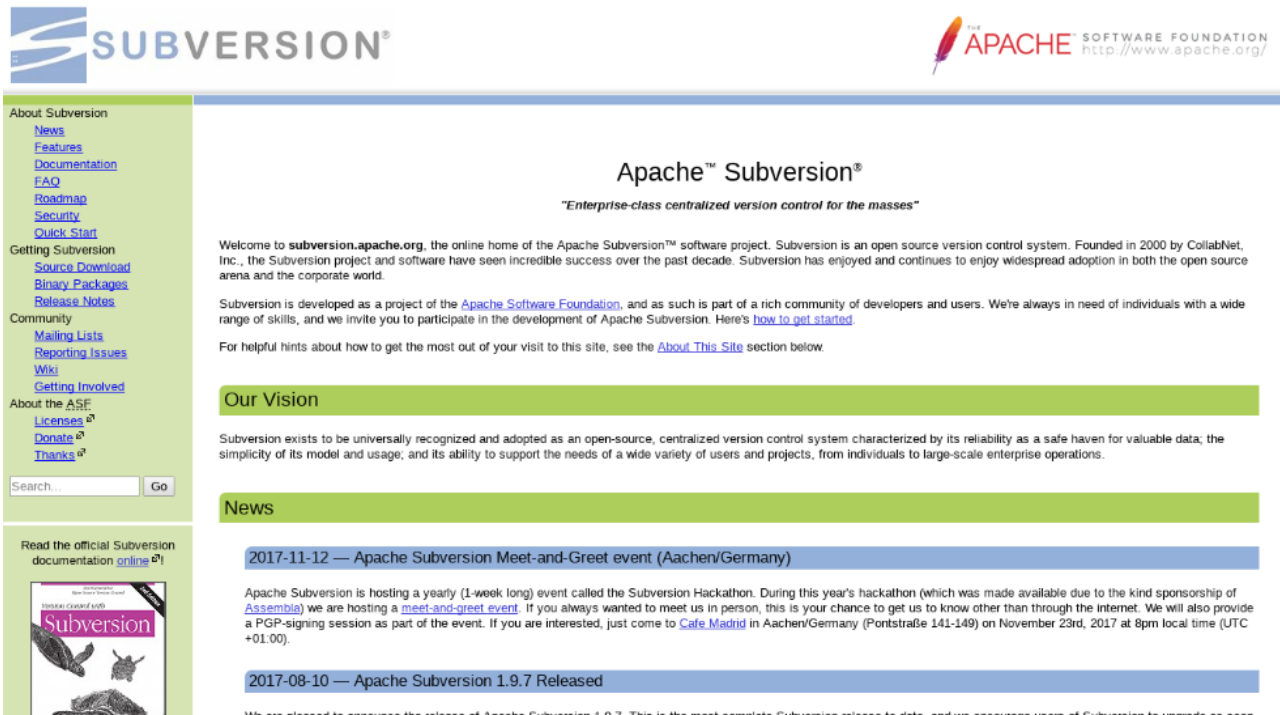
Developer's Blog

- 19-Jan-2012: [Improved quilt patch handling](#)
- 13-Jan-2012: [Extracting and processing SCM data with bzt-xmloutput](#)
- 08-Nov-2011: [What I did on my Rotation](#)

7 Version Control Systems (VCS)

Terdapat banyak Version Control System yang telah dikembangkan para developer. Ada yang berbayar dan ada juga yang free/gratis.

[2] Subversion (SVN) adalah free VCS yang didesain mirip dengan CVS dan lebih sederhana. SVN mendukung atomic commits dari sebuah file dan melakukan versioning terhadap direktori, symbolic links, dan meta-data. SVN merupakan bagian dari Apache Software Foundation. SVN bersifat open source yang didirikan pada tahun 2000 oleh CollabNet Inc. SVN dinikmati luas oleh kalangan komunitas ataupun enterprise. SVN tersedia untuk Linux, OS X, FreeBSD, dan Windows. Saat ini SVN berada di versi 1.9.



The screenshot shows the Apache Subversion website. The header features the Subversion logo on the left and the Apache Software Foundation logo on the right. The main content area is titled "Apache™ Subversion®" with the tagline "Enterprise-class centralized version control for the masses". Below this, a welcome message states that Subversion is an open source version control system founded in 2000 by CollabNet, Inc. A sidebar on the left contains links for "About Subversion", "Getting Subversion", "Community", and "About the ASF". A "News" section at the bottom highlights two events: the "2017-11-12 — Apache Subversion Meet-and-Greet event (Aachen/Germany)" and the "2017-08-10 — Apache Subversion 1.9.7 Released".

Subversion

THE APACHE SOFTWARE FOUNDATION
<http://www.apache.org/>

About Subversion
[News](#)
[Features](#)
[Documentation](#)
[FAQ](#)
[Roadmap](#)
[Security](#)
[Quick Start](#)

Getting Subversion
[Source Download](#)
[Binary Packages](#)
[Release Notes](#)

Community
[Mailing Lists](#)
[Reporting Issues](#)
[Wiki](#)
[Getting Involved](#)

About the ASF
[Licenses](#)
[Donate](#)
[Thanks](#)

Search...

Read the official Subversion documentation [online](#)

Apache™ Subversion®
"Enterprise-class centralized version control for the masses"

Welcome to subversion.apache.org, the online home of the Apache Subversion™ software project. Subversion is an open source version control system. Founded in 2000 by CollabNet, Inc., the Subversion project and software have seen incredible success over the past decade. Subversion has enjoyed and continues to enjoy widespread adoption in both the open source arena and the corporate world.

Subversion is developed as a project of the [Apache Software Foundation](#), and as such is part of a rich community of developers and users. We're always in need of individuals with a wide range of skills, and we invite you to participate in the development of Apache Subversion. Here's [how to get started](#).

For helpful hints about how to get the most out of your visit to this site, see the [About This Site](#) section below.

Our Vision

Subversion exists to be universally recognized and adopted as an open-source, centralized version control system characterized by its reliability as a safe haven for valuable data; the simplicity of its model and usage; and its ability to support the needs of a wide variety of users and projects, from individuals to large-scale enterprise operations.

News

2017-11-12 — Apache Subversion Meet-and-Greet event (Aachen/Germany)

Apache Subversion is hosting a yearly (1-week long) event called the Subversion Hackathon. During this year's hackathon (which was made available due to the kind sponsorship of [Assemble](#)) we are hosting a [meet-and-greet event](#). If you always wanted to meet us in person, this is your chance to get us to know other than through the internet. We will also provide a PGP-signing session as part of the event. If you are interested, just come to [Cafe Madrid](#) in Aachen/Germany (Pontstraße 141-149) on November 23rd, 2017 at 8pm local time (UTC +01:00).

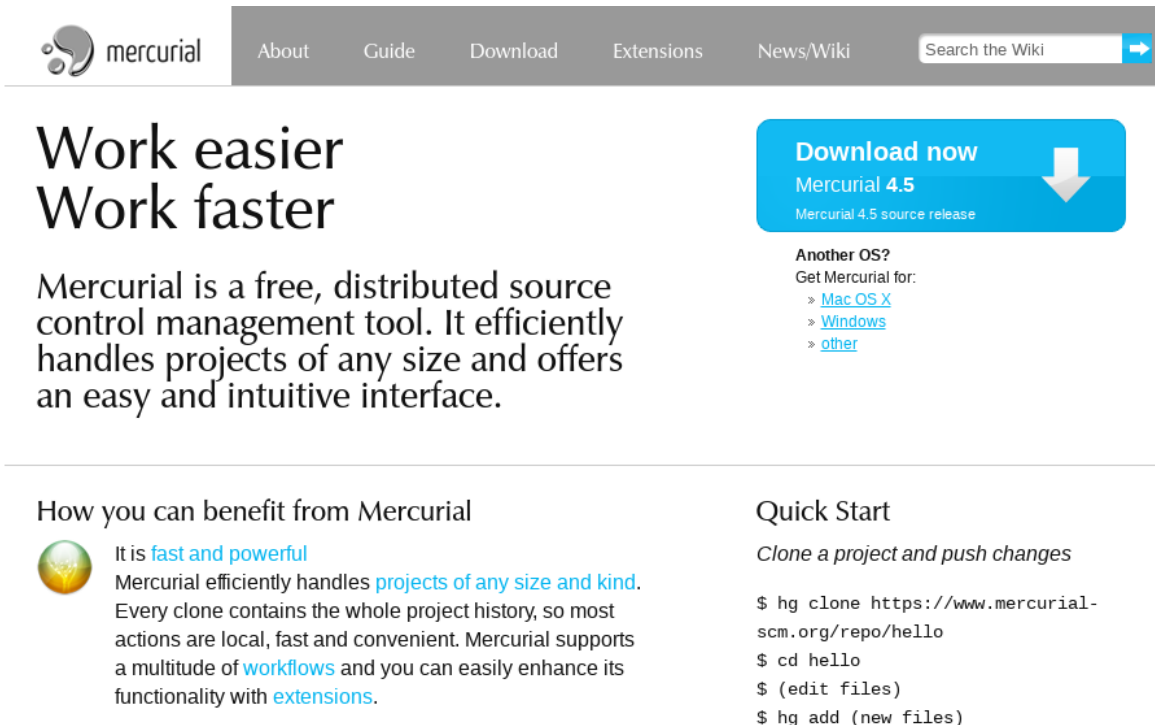
2017-08-10 — Apache Subversion 1.9.7 Released

We are pleased to announce the release of Apache Subversion 1.9.7. This is the most complete Subversion release to date, and we encourage users of Subversion to upgrade as soon

7 Version Control Systems (VCS)

Terdapat banyak Version Control System yang telah dikembangkan para developer. Ada yang berbayar dan ada juga yang free/gratis.

[3] Mercurial merupakan version control system terdistribusi berbasis open source. Mercurial dirancang untuk proyek yang lebih besar, kemungkinan besar berada di luar lingkup designer dan independen Web Developer. Itu tidak berarti tim developer kecil tidak bisa atau tidak boleh menggunakannya. Mercurial memiliki konsep distributed source control management tool. Efisien dalam menangani proyek yang memiliki bermacam ukuran dan jenis. Mercurial mendukung berbagai workflow.



The image is a screenshot of the Mercurial website. At the top, there is a navigation bar with links: 'About', 'Guide', 'Download', 'Extensions', 'News/Wiki', and a search box labeled 'Search the Wiki'. Below the navigation bar, the main content area features the text 'Work easier Work faster' in large, bold letters. To the right of this text is a blue button that says 'Download now' with a downward arrow, and below it, 'Mercurial 4.5' and 'Mercurial 4.5 source release'. Further down, there is a section titled 'Another OS?' with links for 'Mac OS X', 'Windows', and 'other'. Below this, there is a section titled 'How you can benefit from Mercurial' with a small globe icon and text describing its features. To the right of this section is a 'Quick Start' section with a list of commands to clone a project and push changes.

mercurial

About Guide Download Extensions News/Wiki Search the Wiki

Work easier Work faster

Mercurial is a free, distributed source control management tool. It efficiently handles projects of any size and offers an easy and intuitive interface.

Download now
Mercurial 4.5
Mercurial 4.5 source release

Another OS?
Get Mercurial for:
> [Mac OS X](#)
> [Windows](#)
> [other](#)

How you can benefit from Mercurial

It is [fast and powerful](#)
Mercurial efficiently handles [projects of any size and kind](#). Every clone contains the whole project history, so most actions are local, fast and convenient. Mercurial supports a multitude of [workflows](#) and you can easily enhance its functionality with [extensions](#).

Quick Start

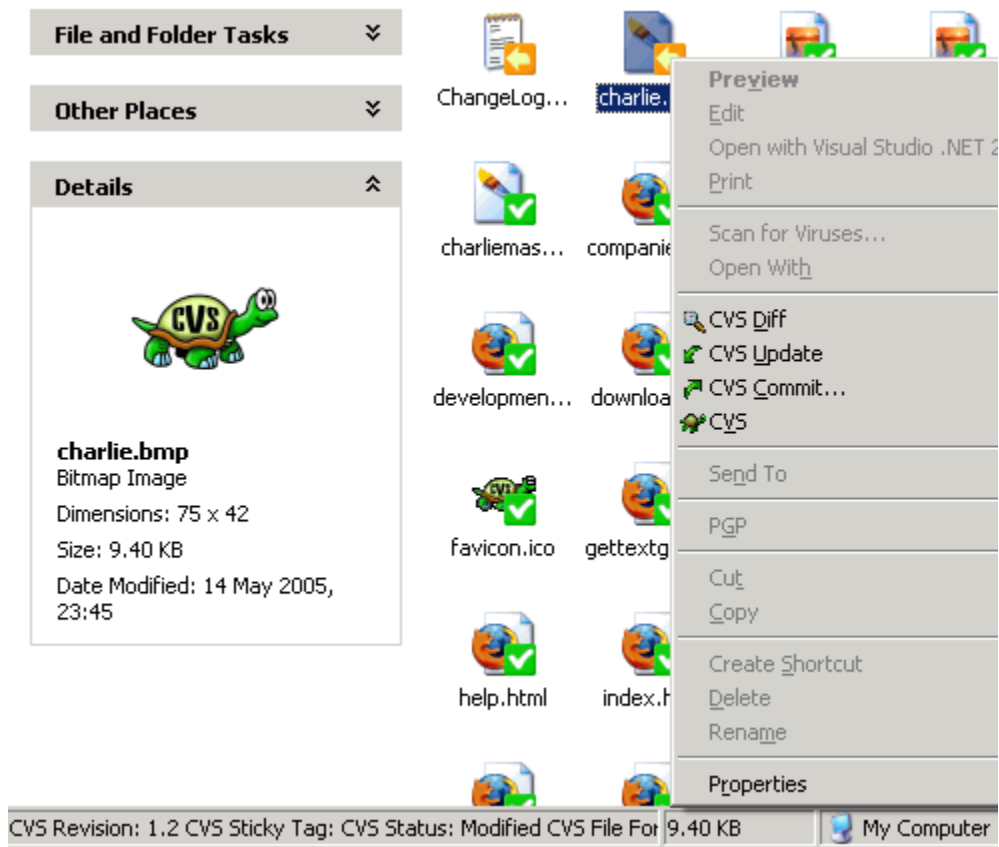
Clone a project and push changes

```
$ hg clone https://www.mercurial-scm.org/repo/hello
$ cd hello
$ (edit files)
$ hg add (new files)
```

7 Version Control Systems (VCS)

Terdapat banyak Version Control System yang telah dikembangkan para developer. Ada yang berbayar dan ada juga yang free/gratis.

[4] **CVS** adalah revision control systems yang pertama kali dibuat. pertama kali dirilis pada tahun 1986, dan Google Code masih menjadi tuan rumah posting Usenet yang asli yang mengumumkan CVS.



7 Version Control Systems (VCS)

Terdapat banyak Version Control System yang telah dikembangkan para developer. Ada yang berbayar dan ada juga yang free/gratis.

[5] Revision Control System (RCS) mampu mengelola revisi dari banyak file. RCS mengotomasi penyimpanan, pengambilan, pencatatan, pemeriksaan, dan penggabungan dari sebuah revisi. Berguna untuk teks yang sering direvisi seperti source code, program, dokumentasi, grafik, jurnal, dan surat.

GNU RCS

The Revision Control System (RCS) manages multiple revisions of files. RCS automates the storing, retrieval, logging, identification, and merging of revisions. RCS is useful for text that is revised frequently, including source code, programs, documentation, graphics, papers, and form letters.

RCS was first developed by Walter F. Tichy at Purdue University in the early 1980s -- paper: [RCS: A System for Version Control \(1991\)](#) (troff, PostScript, PDF). See also the [Purdue RCS Homepage](#).

RCS design is an improvement from its predecessor Source Code Control System (SCCS) (see [GNU CSSC](#)). The improvements include an easier user interface and improved storage of versions for faster retrieval. RCS improves performance by storing an entire copy of the most recent version and then stores *reverse* differences (called "deltas"). RCS uses [GNU Diffutils](#) to find the differences between versions.

Download / News

([FTP mirrors](#))

Latest release: [5.9.4 \(2015-01-22\)](#)

- portability fix in "make check" for OSX

We now avoid 'head -N', where N is a number, since that construct is not portable. See: <http://lists.gnu.org/archive/html/bug-rcs/2014-11/msg00000.html>

- doc improvements
 - index 'rcs -o' better

It seems the term "outdate" is itself outdated, nowadays (sigh). This command is now indexed under "deleting" and "removing", as well as "outdating".

7 Version Control Systems (VCS)

Terdapat banyak Version Control System yang telah dikembangkan para developer. Ada yang berbayar dan ada juga yang free/gratis.

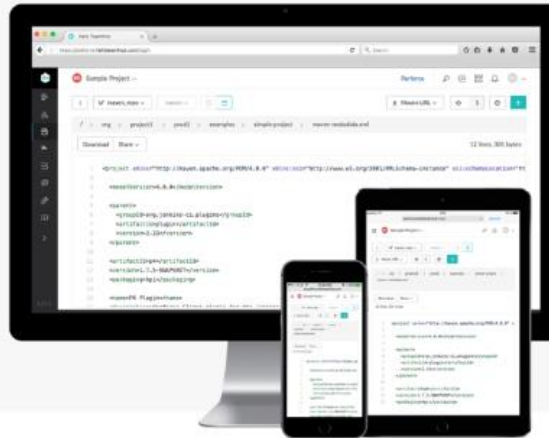
[6] Perforce dipercaya oleh vendor-vendor besar seperti Netflix, Samsung, Salesforce, dan New York Stock Exchange. Perforce memiliki keunggulan seperti massive scalability, hybrid version control, social coding, large binaries, dan unified security. Perforce dapat digunakan di Linux, varian Unix, OS X, dan Windows. Saat ini Perforce berada di versi 2017.3 untuk versi Free.

New from Perforce!



Git code hosting and collaboration.
Unparalleled CI/CD performance for DevOps.

TRY IT NOW



400,000+ Users Strong

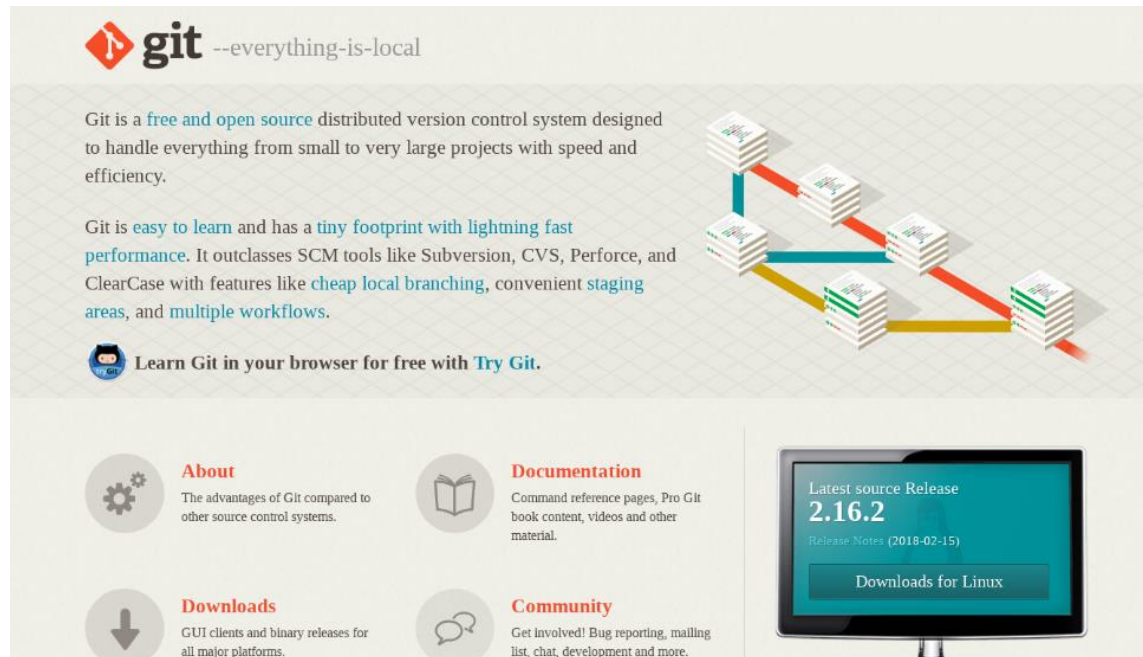
Leaders across all industries trust their success to Perforce.



7 Version Control Systems (VCS)

Terdapat banyak Version Control System yang telah dikembangkan para developer. Ada yang berbayar dan ada juga yang free/gratis.

[7] **Git** merupakan Version Control Sistem yang dikembangkan oleh Linux Torvald ketika mengembangkan kernel Linux pada tahun 2005. Git mengizinkan dan mendorong kita untuk memiliki beberapa cabang lokal yang dapat sepenuhnya independen satu sama lain. Pembuatan, penggabungan, dan penghapusan jalur pengembangan membutuhkan waktu dalam beberapa detik. Git mempunyai keunggulan seperti repository syncing, bekerja secara offline, cheap local branching, staging area yang nyaman, mampu menangani proyek besar seperti Kernel Linux secara efektif dalam hal kecepatan dan ukuran data, mendukung non-linear development, dan multiple workflow.



The image shows the Git website banner and navigation menu. The banner features the Git logo (a red diamond with a white 'T' and the word 'git' in lowercase) followed by the tagline '--everything-is-local'. Below this, there is a paragraph describing Git as a free and open source distributed version control system. To the right of the text is a diagram showing a branching model with several stacks of books representing code repositories, connected by colored lines (red, blue, yellow) representing branches and merges. Below the banner is a navigation menu with four items: 'About' (with a gear icon), 'Documentation' (with a book icon), 'Downloads' (with a download arrow icon), and 'Community' (with a speech bubble icon). Each item has a brief description. To the right of the navigation menu is a monitor displaying the latest source release '2.16.2' and a button for 'Downloads for Linux'.

git --everything-is-local

Git is a [free and open source](#) distributed version control system designed to handle everything from small to very large projects with speed and efficiency.

Git is [easy to learn](#) and has a [tiny footprint with lightning fast performance](#). It outclasses SCM tools like Subversion, CVS, Perforce, and ClearCase with features like [cheap local branching](#), convenient [staging areas](#), and [multiple workflows](#).

 Learn Git in your browser for free with [Try Git](#).

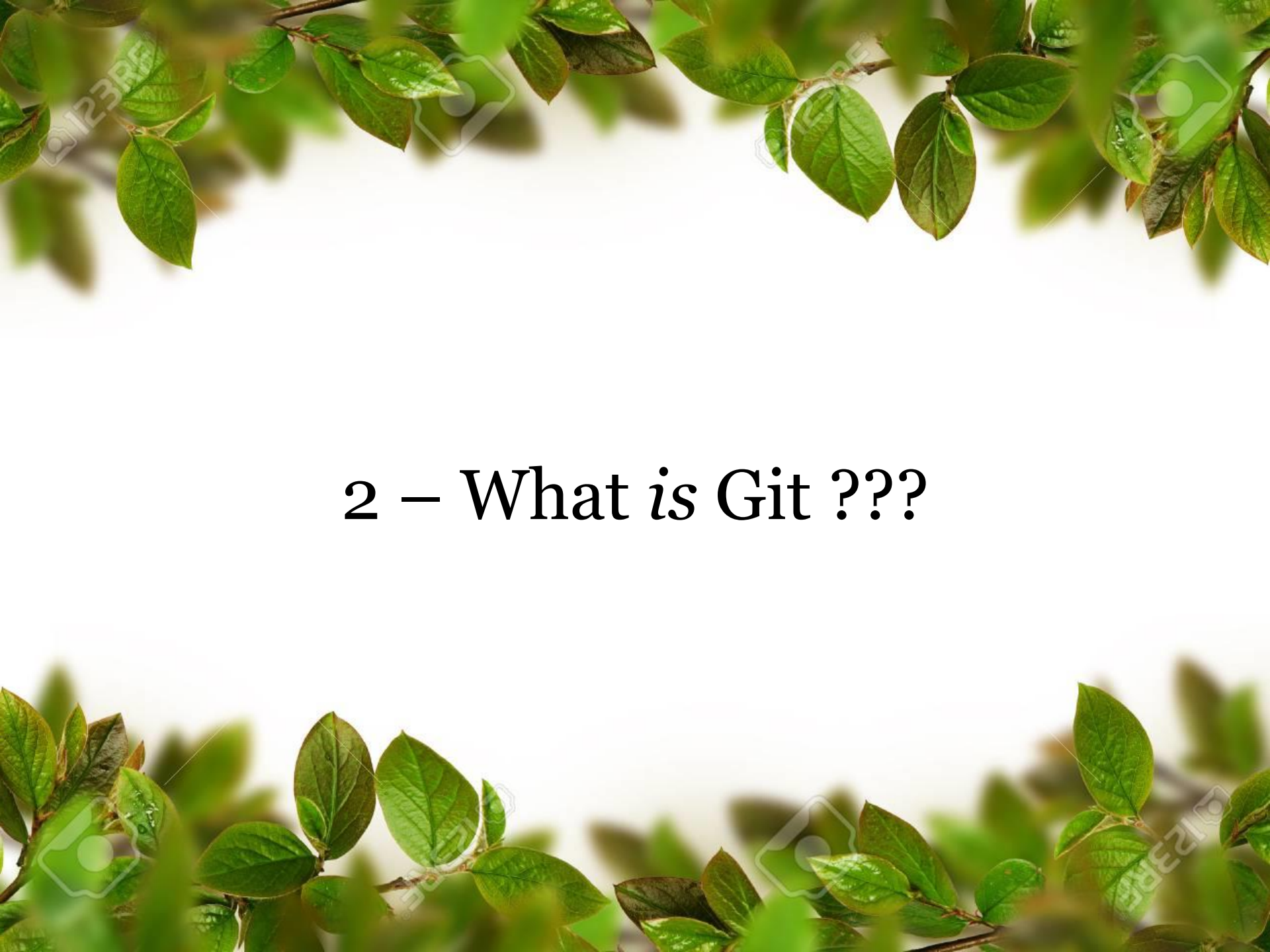
About
The advantages of Git compared to other source control systems.

Documentation
Command reference pages, Pro Git book content, videos and other material.

Downloads
GUI clients and binary releases for all major platforms.

Community
Get involved! Bug reporting, mailing list, chat, development and more.

Latest source Release
2.16.2
Release Notes (2018-02-15)
[Downloads for Linux](#)



2 – What is Git ???

<https://git-scm.com/book/id/v2/Memulai-Sejarah-Singkat-Git>

The screenshot shows the Git website with the title "git --fast-version-control". The left sidebar contains links for "About", "Documentation" (with sub-links: Reference, Book, Videos, External Links), "Downloads", and "Community". The main content area is titled "1.2 Memulai - Sejarah Singkat Git" and includes a search bar. The text discusses the history of Git, mentioning its origin in the Linux kernel project and its development by Linus Torvalds. It also lists several languages in which the book is available, including English, Azerbaijani, Bulgarian, German, Spanish, French, Greek, Japanese, Korean, Dutch, and Russian.

1.2 Memulai - Sejarah Singkat Git

Sejarah Singkat Git

Bersamaan dengan banyak hal besar dalam hidup, Git dimulai dengan sedikit kehancuran kreatifitas dan pertentangan yang ganas.

Kernel Linux adalah proyek perangkat lunak sumber terbuka dalam lingkup yang cukup besar. Sebagian besar waktu pemeliharaan dari kernel Linux (1991-2002), perubahan-perubahan pada perangkat lunak diberikan sebagai patch dan berkas terarsipkan. Pada 2002, proyek kernel Linux mulai menggunakan DVCS terpatenkan bernama BitKeeper.

Pada 2005, hubungan antara komunitas yang mengembangkan BitKeeper terputus, dan mendesak komunitas pengembangan Linux (kt mengembangkan alat mereka sendiri berdasar ketika menggunakan BitKeeper. Beberapa sas

- Kecepatan
- Rancangan yang sederhana
- Dukungan yang kuat untuk pengembangan
- Benar-benar tersebar
- Mampu menangani proyek besar seperti

Tutorial Git & Github ubuntu.

The screenshot shows the Git website with the title "git --fast-version-control". The left sidebar contains links for "About", "Documentation" (with sub-links: Reference, Book, Videos, External Links), "Downloads", and "Community". The main content area is titled "1.2 Memulai - Sejarah Singkat Git" and includes a search bar. The text discusses the history of Git, mentioning its origin in the Linux kernel project and its development by Linus Torvalds. It also lists several languages in which the book is available, including English, Azerbaijani, Bulgarian, German, Spanish, French, Greek, Japanese, Korean, Dutch, and Russian.

1.2 Memulai - Sejarah Singkat Git

Sejarah Singkat Git

Bersamaan dengan banyak hal besar dalam hidup, Git dimulai dengan sedikit kehancuran kreatifitas dan pertentangan yang ganas.

Kernel Linux adalah proyek perangkat lunak sumber terbuka dalam lingkup yang cukup besar. Sebagian besar waktu pemeliharaan dari kernel Linux (1991-2002), perubahan-perubahan pada perangkat lunak diberikan sebagai patch dan berkas terarsipkan. Pada 2002, proyek kernel Linux mulai menggunakan DVCS terpatenkan bernama BitKeeper.

Pada 2005, hubungan antara komunitas yang mengembangkan BitKeeper terputus, dan mendesak komunitas pengembangan Linux (kt mengembangkan alat mereka sendiri berdasar ketika menggunakan BitKeeper. Beberapa sas

- Kecepatan
- Rancangan yang sederhana
- Dukungan yang kuat untuk pengembangan
- Benar-benar tersebar
- Mampu menangani proyek besar seperti

What is Git?

- 1) Git adalah **Version Control System (VCS)** tool yang digunakan para developer untuk mengembangkan software secara bersama-bersama yang diciptakan oleh Linus Torvalds.
- 2) Fungsi utama **git** yaitu mengatur versi dari source code program dengan memberikan tanda baris dan code mana yang ditambah atau diganti.
- 3) Git memudahkan *programmer* untuk mengetahui perubahan source code daripada harus membuat file baru seperti **Program.java**, **ProgramRevisi.java**, ProgramRevisi2.java, ProgramFix.java.
- 4) Dengan *git* tidak perlu khawatir code yang dikerjakan bentrok, karena setiap developer bisa membuat *branch* sebagai workspace-nya.
- 5) Selain itu, pada **git** bisa memberi komentar pada *source code* yang telah ditambah/diubah, hal ini mempermudah developer lain untuk mengetahui kendala apa yang dialami developer lain.

Distributed Revision Control

- 1) Pengontrol versi pada Git bertugas **mencatat setiap perubahan pada file proyek** yang dikerjakan oleh banyak orang maupun sendiri.
- 2) Git dikenal juga dengan **distributed revision control (VCS terdistribusi)**, artinya penyimpanan database Git tidak hanya berada dalam satu tempat saja.
- 3) **Semua orang yang terlibat dalam pengkodean proyek akan menyimpan database Git**, sehingga akan memudahkan dalam mengelola proyek baik online maupun offline.
- 4) Dalam Git terdapat merge untuk menyebut aktifitas penggabungan kode.
- 5) Sedangkan pada VCS (Version Control System) yang terpusat database disimpan dalam satu tempat dan setiap perubahan disimpan ke sana.

- ❖ GIT merupakan sebuah Version Control System (VCS) yang digunakan dalam tim pengembangan perangkat lunak untuk bekerja bersama.
- ❖ Version Control maksudnya sistem Git akan mencatat setiap perubahan yang terjadi pada source code sehingga memungkinkan untuk **mengambil kembali source code lama jika suatu saat diinginkan kembali ke versi berapapun** dari aplikasi yang pernah kita tulis.



[IMPORTANT] Git Commands

Git init	untuk membuat <i>repository</i> pada file lokal yang nantinya ada folder <code>.git</code>
Git status	untuk mengetahui status dari <u>repository local</u> .
Git add	menambahkan file baru pada repository yang dipilih.
Git commit	untuk menyimpan perubahan yang dilakukan, tetapi tidak ada perubahan pada <u>remote repository</u> .
Git push	untuk mengirimkan perubahan file setelah di <i>commit</i> ke <i>remote repository</i> .
Git branch	untuk melihat seluruh <i>branch</i> yang ada pada <i>repository</i> .
Git checkout	untuk menukar <i>branch</i> yang aktif dengan <i>branch</i> yang dipilih.
Git merge	untuk menggabungkan <i>branch</i> yang aktif dan <i>branch</i> yang dipilih.
Git clone	membuat salinan/copy repository local.

By understanding **Git**, **Git commands**, and **GitHub**, we can more efficiently [manage our source code](#), [collaborate with our team](#), and [participate in open-source projects](#). It can also help us become a more productive and competitive software developer in the IT industry.



Cara Install Git dan Git Bash di Windows:

<https://youtu.be/ligsdfdR7vI>

Cara Upload Folder dan File ke Github:

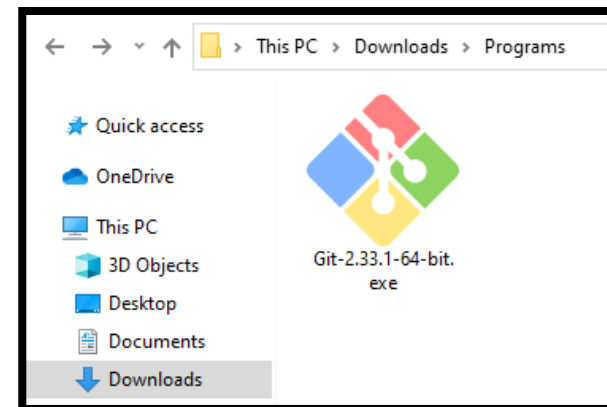
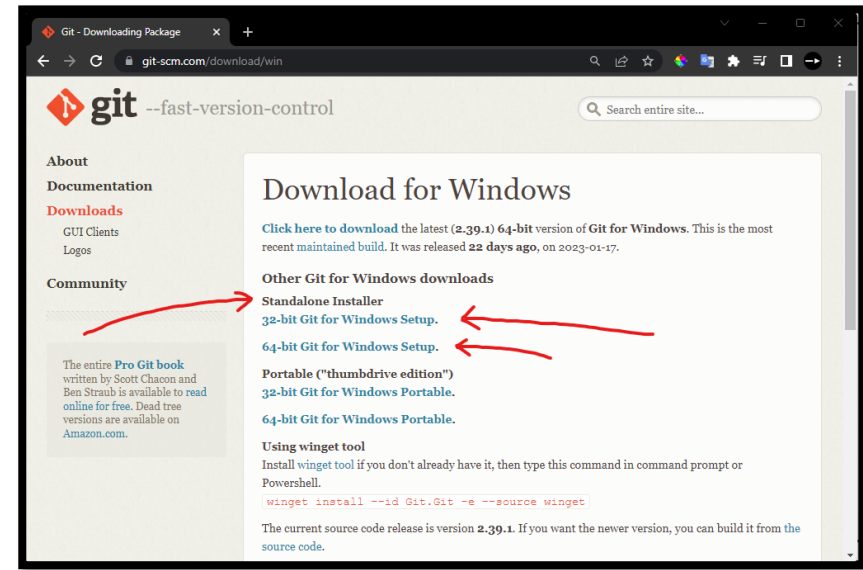
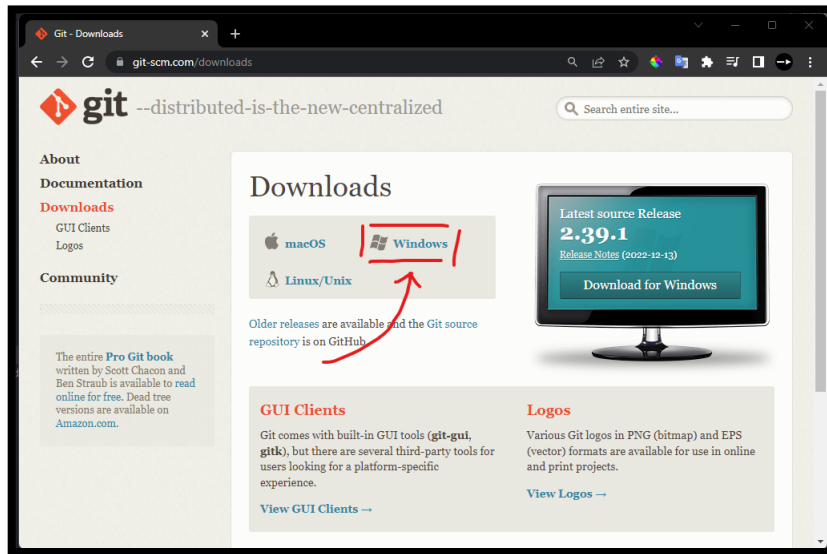
<https://youtu.be/WRsK4f9fmY8>

The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green and appear to be from a tree or shrub. The text is centered in the middle of the slide.

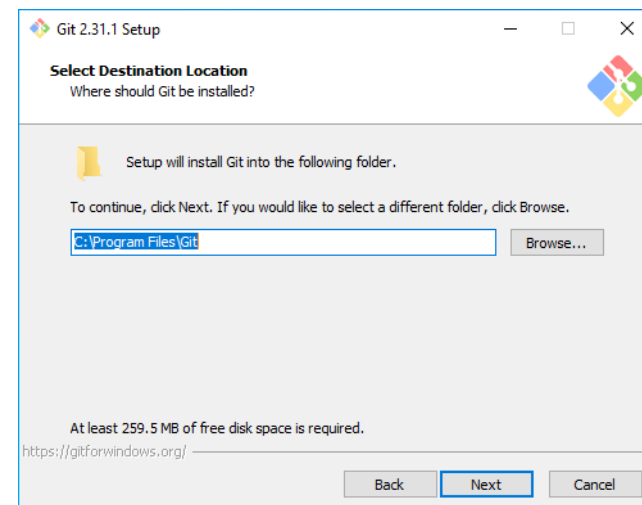
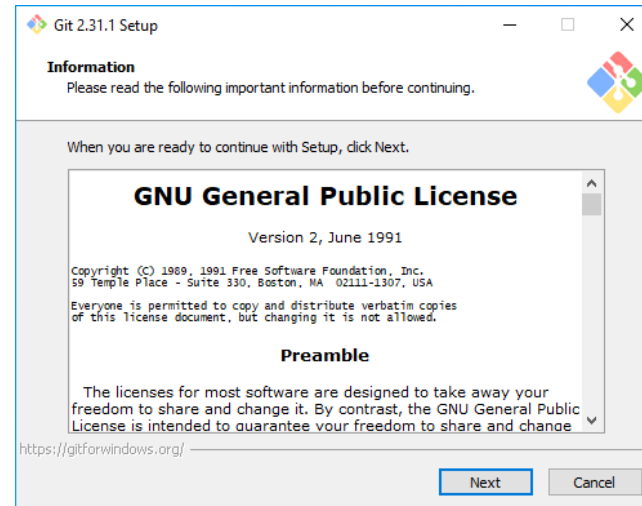
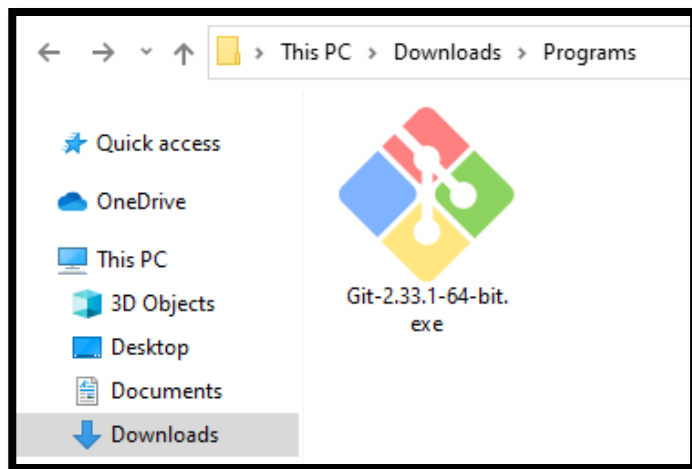
3 – Install Git *on* Windows

Download Installer GIT

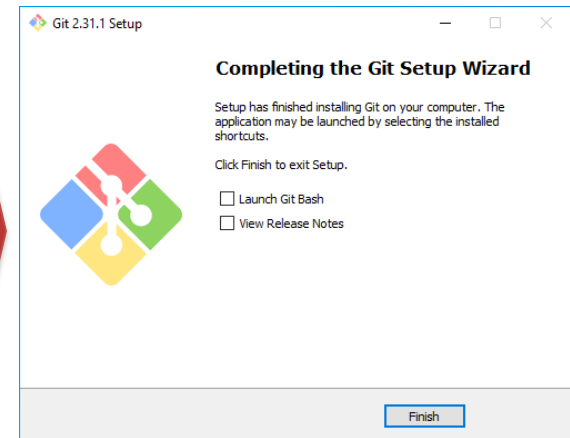
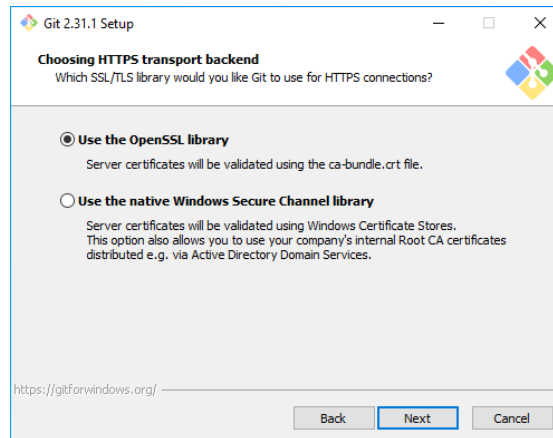
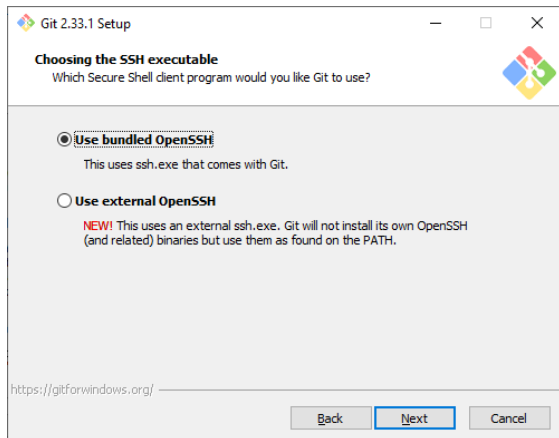
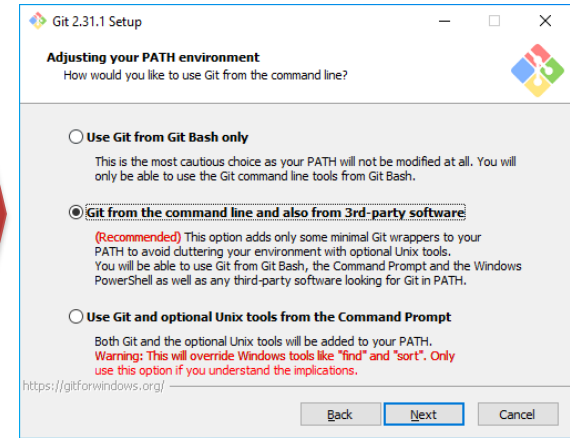
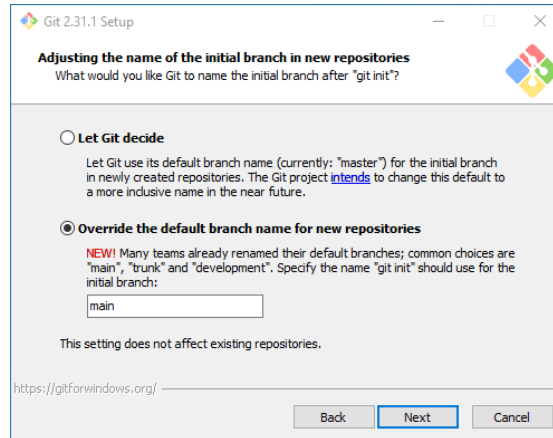
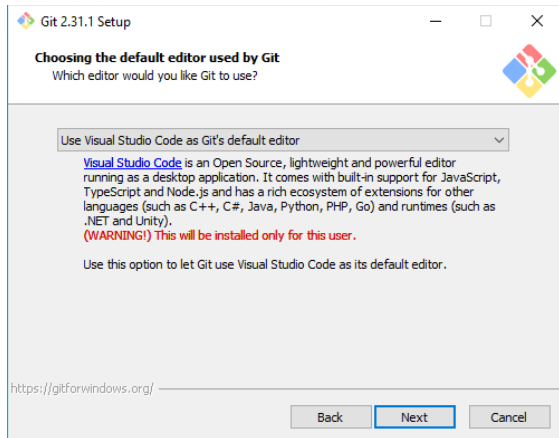
Download Link: <https://git-scm.com/downloads>



Run Git Installer



Konfigurasi GIT Windows



Tutorial Link: <https://aantamim.id/git-windows-install/>

How to Run Git on Windows?

Git memiliki **DUA MODE** yang dapat digunakan, git bash scripting shell (*command line*) dan git graphical user interface (GUI).

[1] Git Bash Scripting Shell (Git Bash)

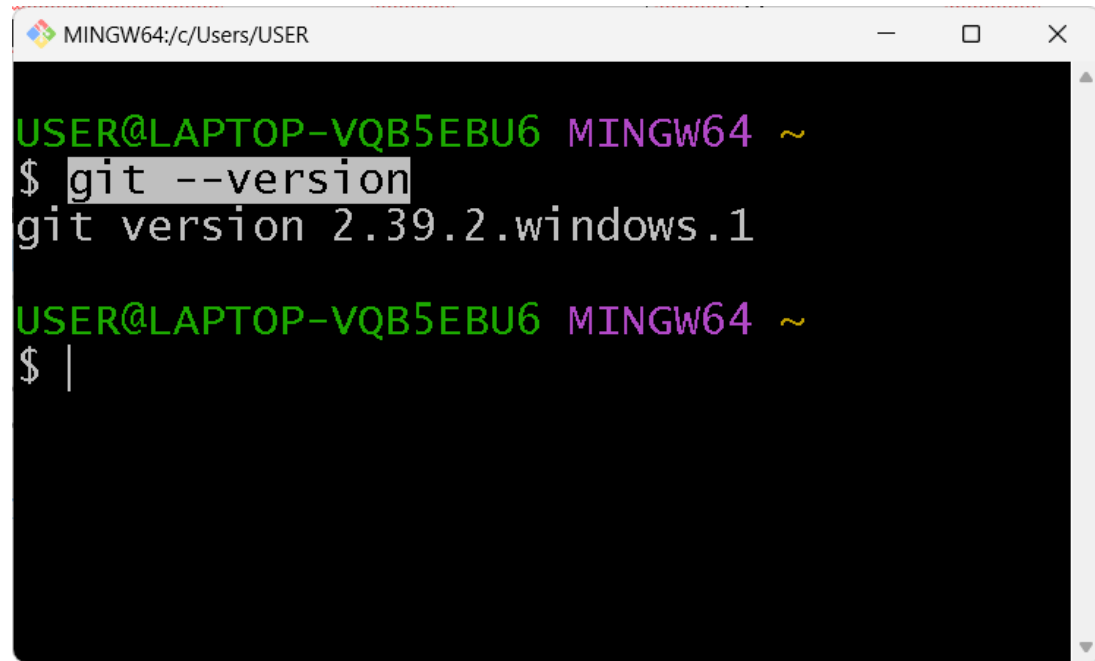
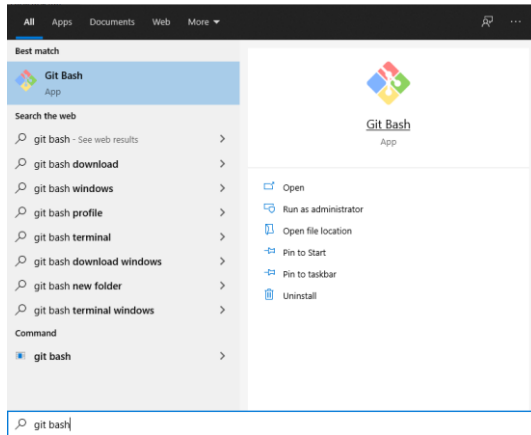
Git Bash adalah sebuah shell atau lingkungan baris perintah yang menjalankan perintah Git. Ini adalah antarmuka berbasis teks di mana kita perlu mengetikkan perintah-perintah Git secara manual. Git Bash tersedia di Windows dan berfungsi seperti lingkungan baris perintah pada sistem operasi Unix.

[2] Git GUI (Graphical User Interface)

Git GUI adalah antarmuka/user interface berbasis grafis yang menyediakan visualisasi dan kontrol yang lebih intuitif untuk Git. Ini menawarkan berbagai alat grafis untuk melakukan operasi Git seperti mengelola cabang, menggabungkan perubahan, mengganti komit, dan banyak lagi. Git GUI tersedia dalam berbagai bentuk dan dapat berbeda antara aplikasi dan platform yang berbeda.

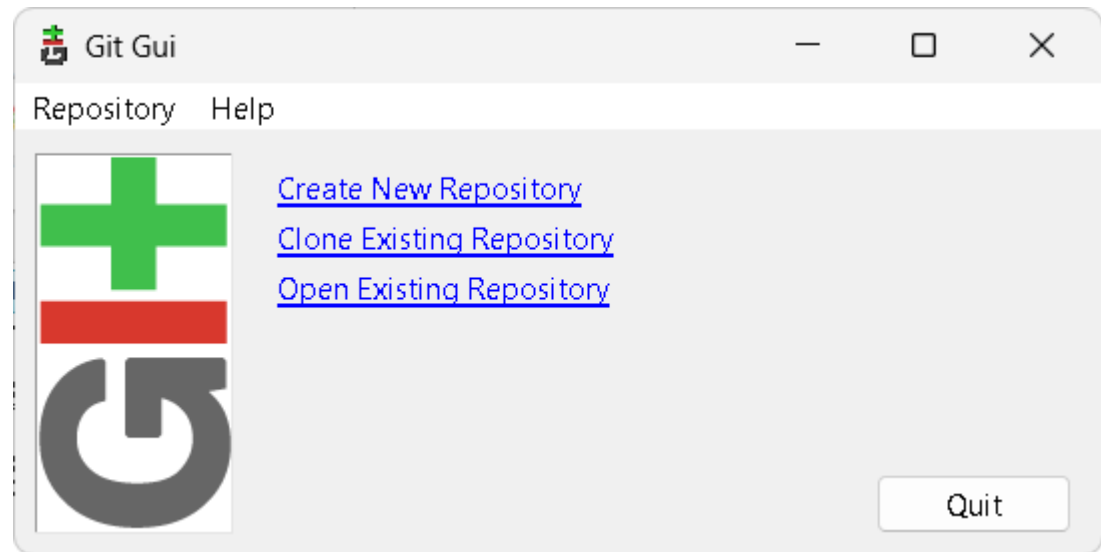
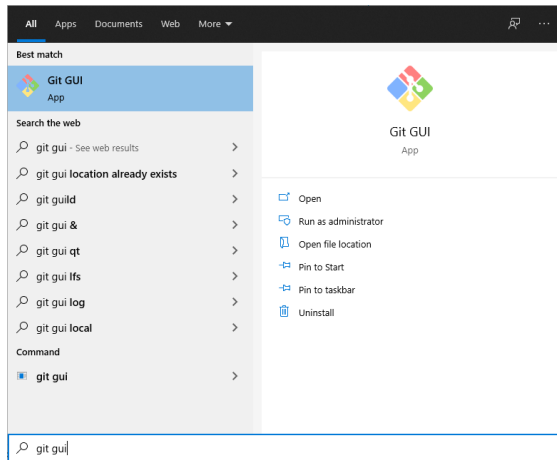
Run/Open “*Git Bash Shell*”

Untuk menjalankan Git Bash buka Start Menu, ketik git bash dan tekan Enter (atau klik dubel pada icon aplikasi).

A screenshot of a Git Bash terminal window. The window title bar shows 'MINGW64:/c/Users/USER'. The terminal prompt is 'USER@LAPTOP-VQB5EBU6 MINGW64 ~'. The user has entered the command '\$ git --version' and the output is 'git version 2.39.2.windows.1'. The prompt is now '\$ |'.

Run/Open “*Git GUI*”

Untuk menjalankan Git GUI buka Start Menu, ketik git gui dan tekan Enter (atau klik dobel pada icon aplikasi).



A decorative border of green leaves and branches at the top of the page.

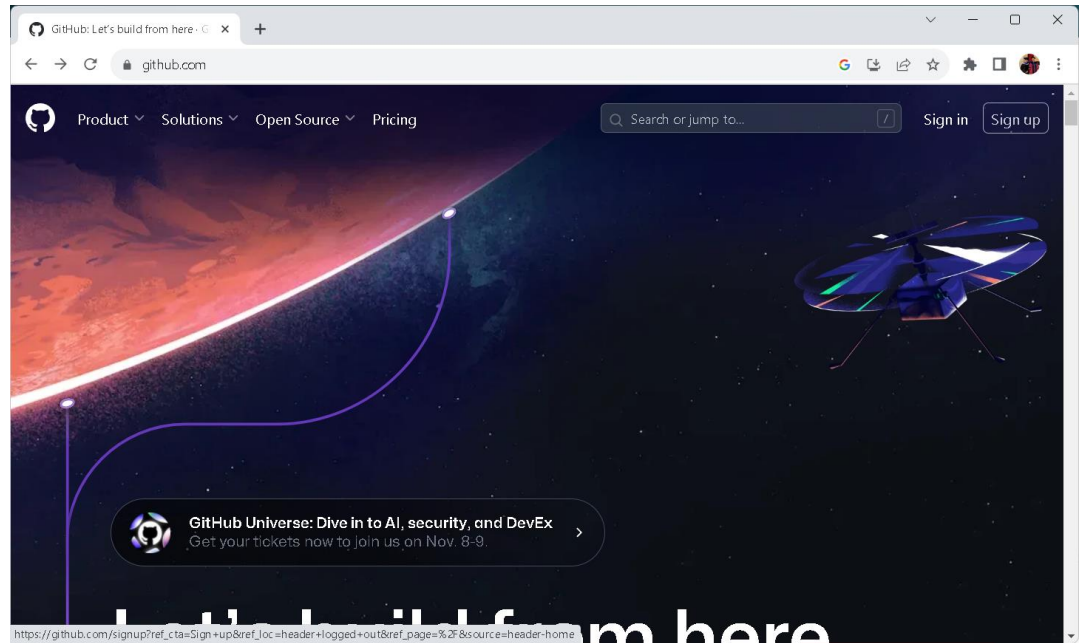
4 – How *to* Create Github Account?

A decorative border of green leaves and branches at the bottom of the page.

Github Link:

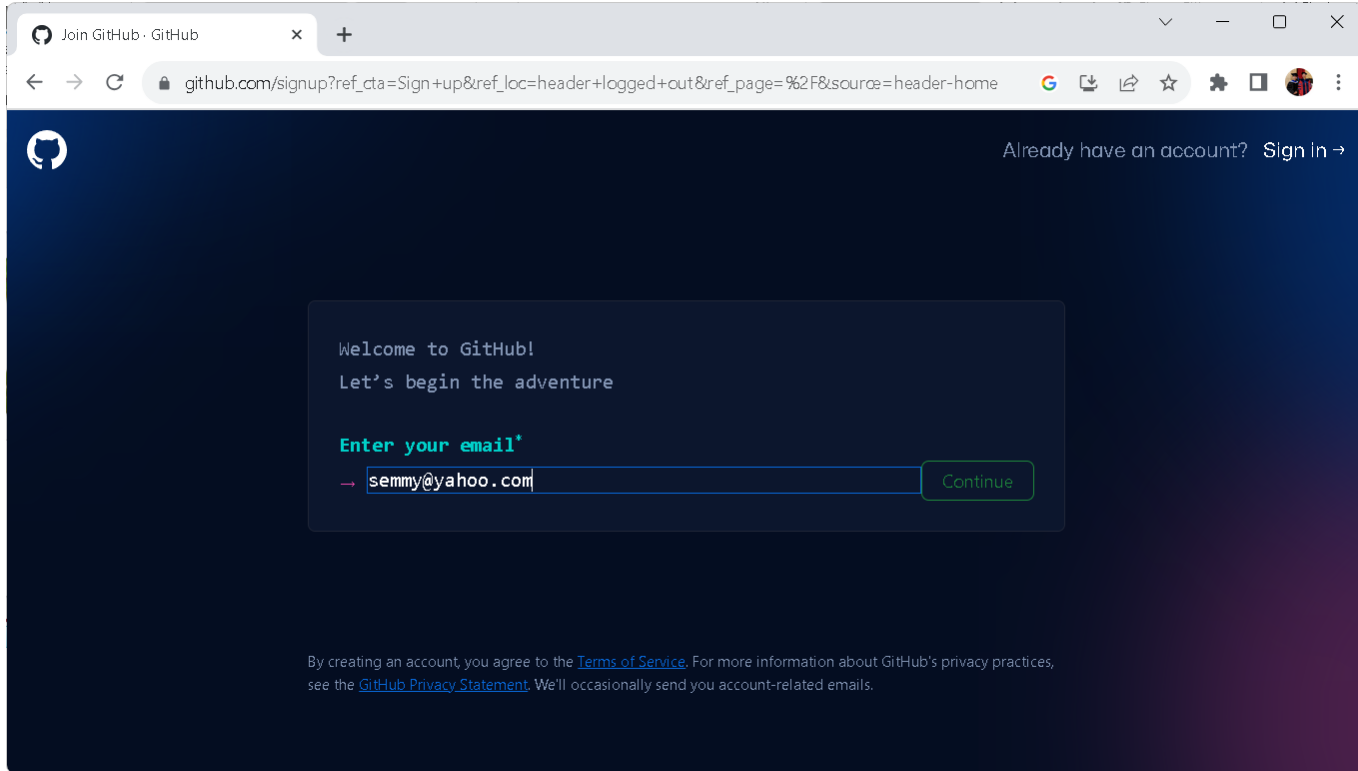
<https://github.com/>

Open Link
Github



Silahkan click button **“Sign Up”** untuk memulai register new account.

Create New Github Account



A screenshot of a web browser showing the GitHub sign-up page. The browser's address bar displays the URL: `github.com/signup?ref_cta=Sign+up&ref_loc=header+logged+out&ref_page=%2F&source=header-home`. The page has a dark blue background with a white GitHub logo in the top left corner. In the top right corner, there is a link that says "Already have an account? Sign In →". The main content area is a dark blue box with white text that reads: "Welcome to GitHub!" followed by "Let's begin the adventure". Below this, there is a prompt "Enter your email*" in green text. A text input field contains the email address "semmy@yahoo.com". To the right of the input field is a green "Continue" button. At the bottom of the page, there is a small line of text: "By creating an account, you agree to the [Terms of Service](#). For more information about GitHub's privacy practices, see the [GitHub Privacy Statement](#). We'll occasionally send you account-related emails."

Join GitHub · GitHub

github.com/signup?ref_cta=Sign+up&ref_loc=header+logged+out&ref_page=%2F&source=header-home

Already have an account? Sign In →

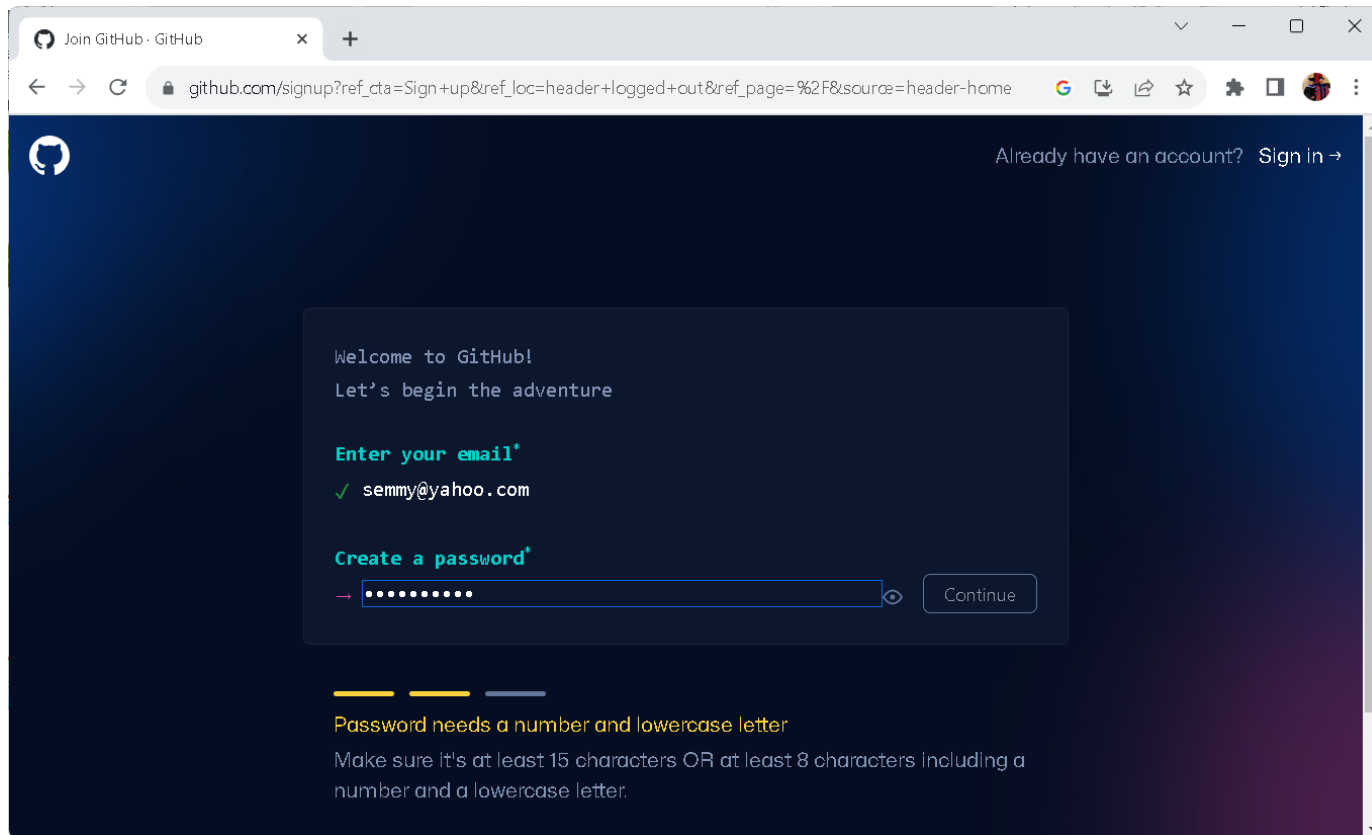
Welcome to GitHub!
Let's begin the adventure

Enter your email*

→ semmy@yahoo.com Continue

By creating an account, you agree to the [Terms of Service](#). For more information about GitHub's privacy practices, see the [GitHub Privacy Statement](#). We'll occasionally send you account-related emails.

Create New Github Account



The screenshot shows the GitHub sign-up page in a web browser. The browser's address bar displays the URL `github.com/signup?ref_cta=Sign+up&ref_loc=header+logged+out&ref_page=%2F&source=header-home`. The page features the GitHub logo in the top left and a link to "Sign in" in the top right. The main content area is a dark blue box with the following text:

Welcome to GitHub!
Let's begin the adventure

Enter your email*
✓ `semmy@yahoo.com`

Create a password*
→ 

[Continue](#)

Below the form, there are three progress bars (two yellow, one grey) and a note: "Password needs a number and lowercase letter". A detailed instruction follows: "Make sure it's at least 15 characters OR at least 8 characters including a number and a lowercase letter."

Join GitHub · GitHub

github.com/signup?ref_cta=Sign+up&ref_loc=header+logged+out&ref_page=%2F&source=header-home

GitHub

Already have an account? [Sign in](#)

Welcome to GitHub!
Let's begin the adventure

Enter your email*

✓ semmy@yahoo.com

Create a password*

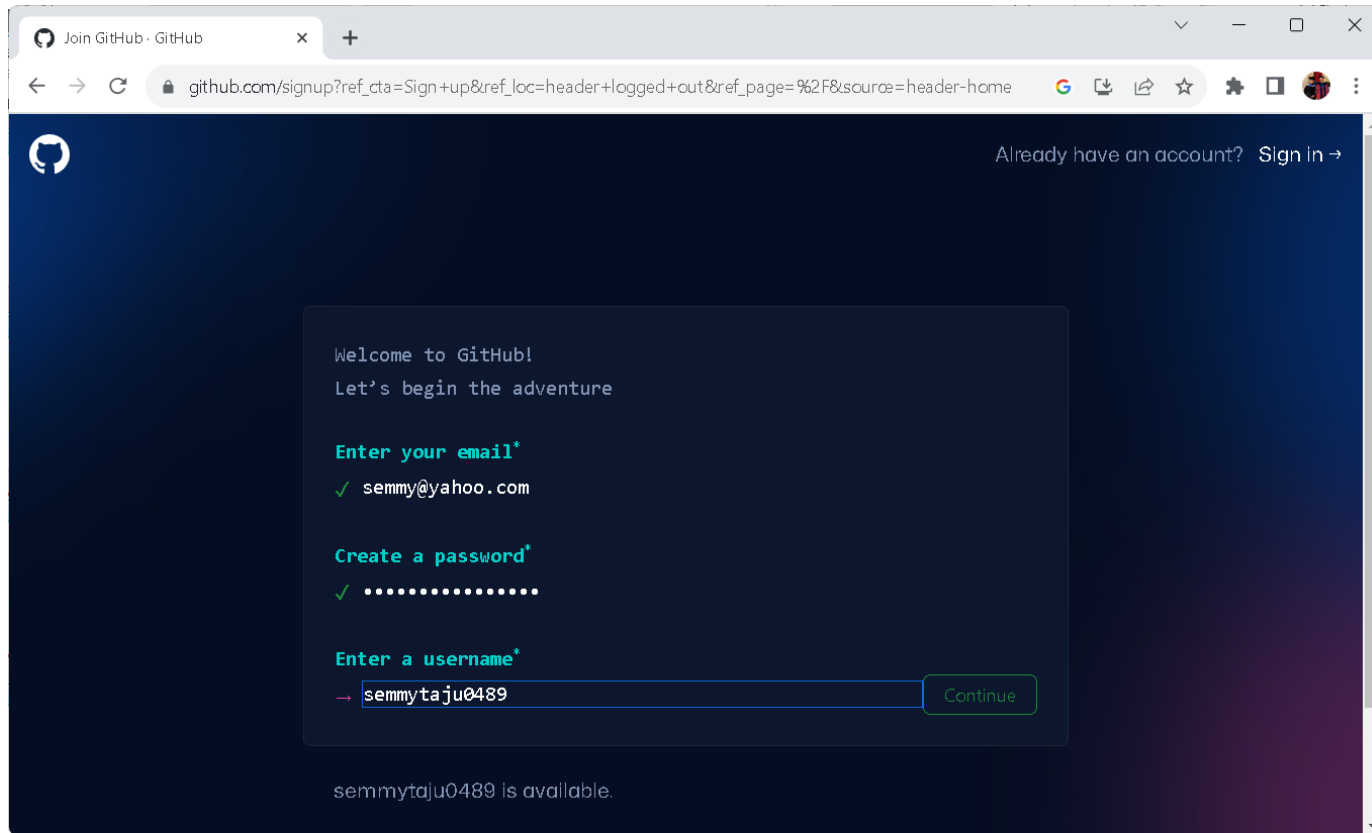
✗ ●●●●●●●●●●●●●●●●

Continue

Password is strong

Make sure it's at least 15 characters OR at least 8 characters including a number and a lowercase letter.

Create UserName



The screenshot shows the GitHub sign-up page in a web browser. The browser's address bar displays the URL: `github.com/signup?ref_cta=Sign+up&ref_loc=header+logged+out&ref_page=%2F&source=header-home`. The page has a dark blue background with the GitHub logo in the top left and a link to "Sign in" in the top right. A central white box contains the following text and form elements:

Welcome to GitHub!
Let's begin the adventure

Enter your email*
✓ `semmy@yahoo.com`

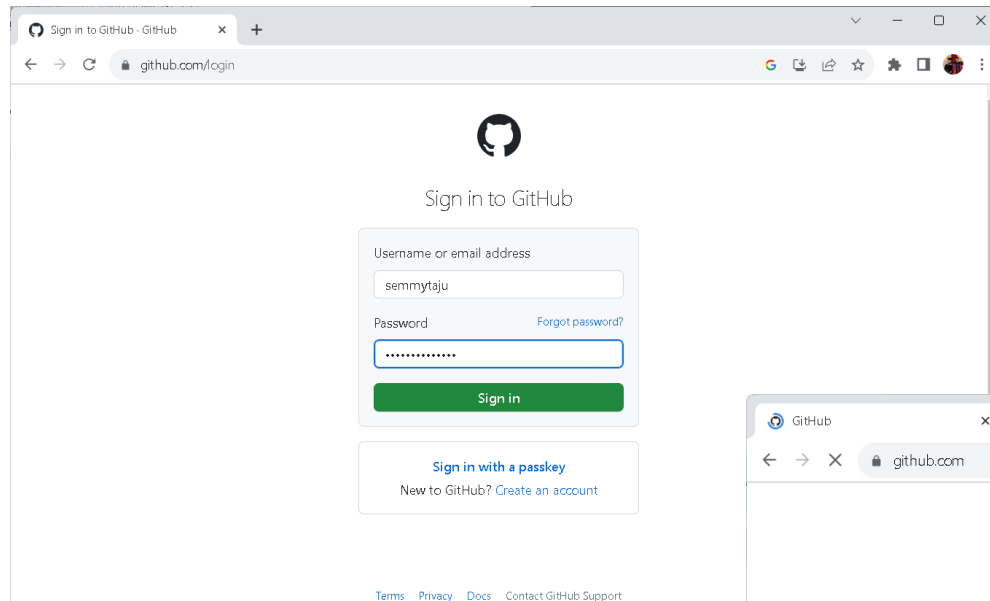
Create a password*
✓ `.....`

Enter a username*
→ `semmytaju0489` Continue

semmytaju0489 is available.

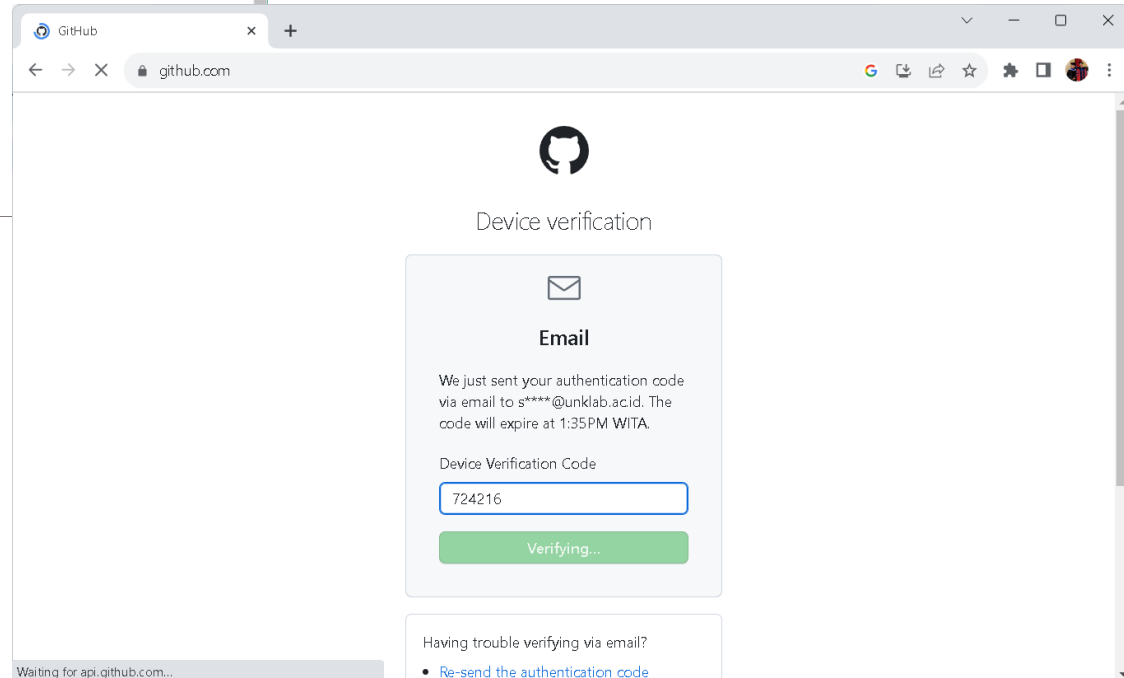
“Sign In” to Github

Github Login: <https://github.com/login>



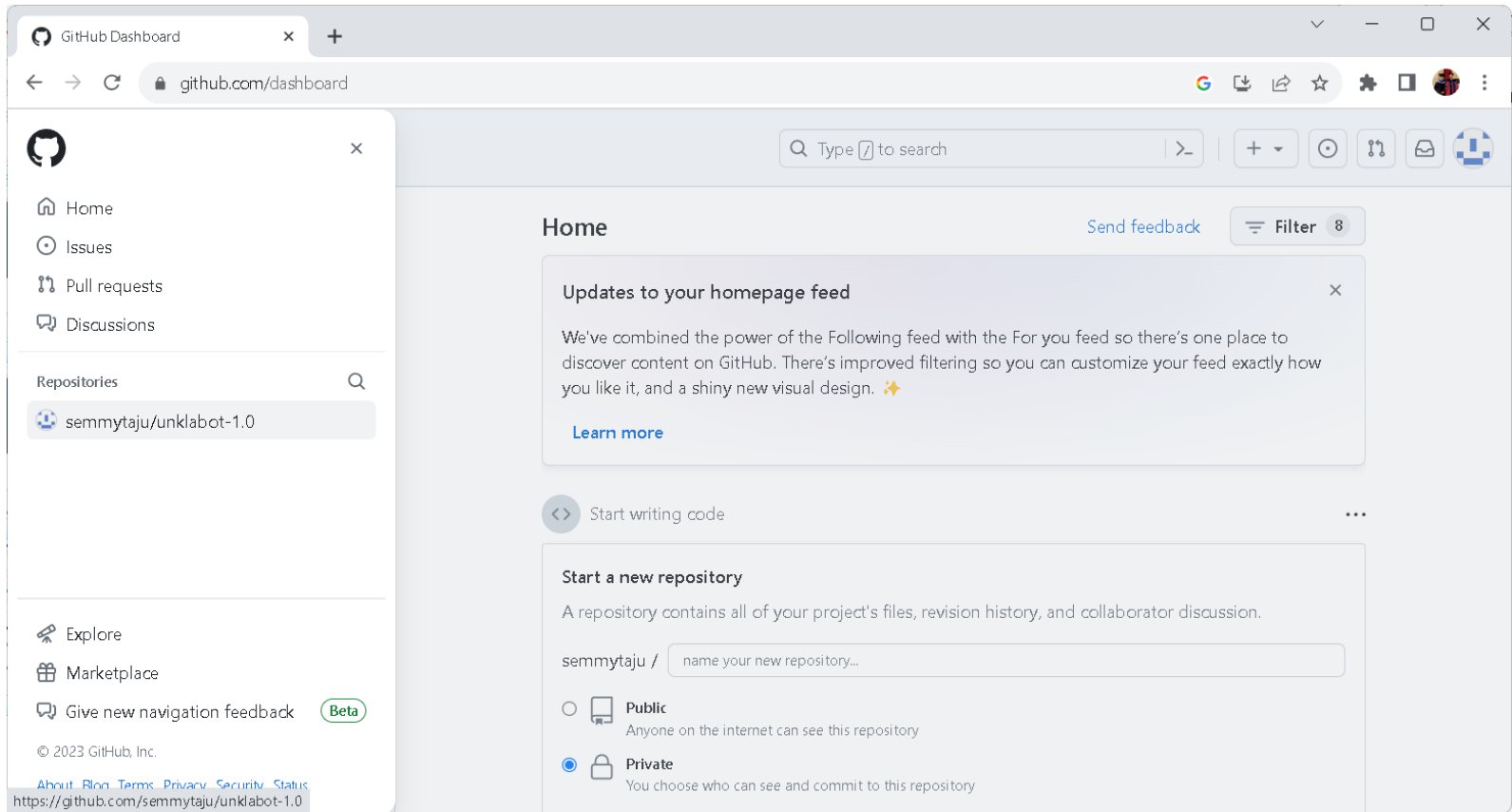
A screenshot of the GitHub login page in a web browser. The browser's address bar shows 'github.com/login'. The page features the GitHub logo at the top, followed by the text 'Sign in to GitHub'. Below this is a login form with two input fields: 'Username or email address' containing 'semmytaju' and 'Password' with masked characters. A 'Forgot password?' link is next to the password field. A green 'Sign in' button is at the bottom of the form. Below the form is a link 'Sign in with a passkey' and a note 'New to GitHub? Create an account'. At the very bottom, there are links for 'Terms', 'Privacy', 'Docs', and 'Contact GitHub Support'.

**Verifikasi
code akan di
kirim ke
email.**



A screenshot of the GitHub device verification page. The browser's address bar shows 'github.com'. The page features the GitHub logo at the top, followed by the text 'Device verification'. Below this is a form with an envelope icon and the heading 'Email'. The text reads: 'We just sent your authentication code via email to s****@unklab.ac.id. The code will expire at 1:35 PM WITA.' Below this is a label 'Device Verification Code' and an input field containing '724216'. A green 'Verifying...' button is at the bottom of the form. At the bottom of the page, there is a link 'Re-send the authentication code' and a status bar at the very bottom that says 'Waiting for api.github.com...'.

Github Dashboard



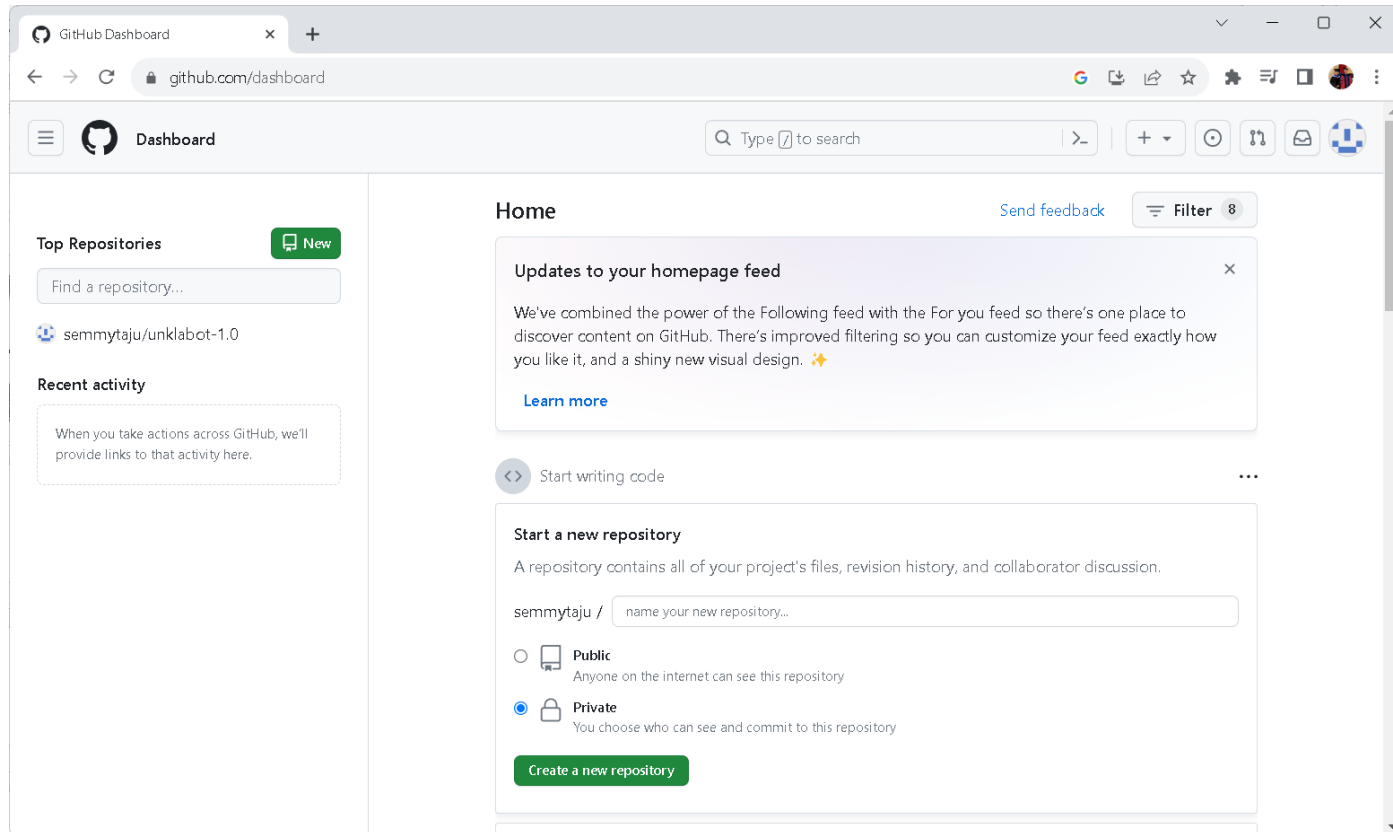


5 – Push Folders & Files to Github

(git push = mengirimkan perubahan file)

Create Repository Github

Kemudian buat repository dengan mengklik tanda + dan pilih **new repository**.



Public Repository Github

Next, masukan nama **repository** cloud/online yang akan dijadikan sebagai tempat code.

Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository](#).

Required fields are marked with an asterisk (*).

Owner *

 semmytaju

Repository name *

front-end-web

✔ front-end-web is available.

Great repository names are short and memorable. Need inspiration? How about [fictional-barnacle](#)?

Description (optional)

☒ Public

Anyone on the internet can see this repository. You choose who can commit.

☐ Private

You choose who can see and commit to this repository.

Initialize this repository with:

☐ Add a README file

This is where you can write a long description for your project. [Learn more about READMEs](#).

Add .gitignore

gitignore template: None

Choose which files not to track from a list of templates. [Learn more about ignoring files](#).

Choose a license

License: None

A license tells others what they can and can't do with your code. [Learn more about licenses](#).

 You are creating a public repository in your personal account.

Create repository

Quick Github Setup

The screenshot shows the GitHub web interface for the repository 'semmytaju / front-end-web'. The browser address bar shows 'github.com/semmytaju/front-end-web'. The repository page includes a header with navigation links (Code, Issues, Pull requests, Actions, Projects, Wiki, Security, Insights, Settings) and repository metadata (Public, Pin, Unwatch, Fork, Star). Below the header, there are two main sections: 'Set up GitHub Copilot' and 'Add collaborators to this repository'. A 'Quick setup' section follows, providing instructions for setting up the repository on a desktop or via command line. The 'Quick setup' section includes a dropdown menu for 'Set up in Desktop' or 'HTTPS' or 'SSH' and a text input field for the repository URL 'https://github.com/semmytaju/front-end-web.git'. Below this, there are three sections for creating a new repository, pushing an existing repository, and importing code from another repository, each with a code block and a copy icon.

semmytaju / front-end-web

github.com/semmytaju/front-end-web

semmytaju / front-end-web

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

front-end-web Public

Pin Unwatch 1 Fork 0 Star 0

Set up GitHub Copilot
Use GitHub's AI pair programmer to autocomplete suggestions as you code.
Get started with GitHub Copilot

Add collaborators to this repository
Search for people using their GitHub username or email address.
Invite collaborators

Quick setup — if you've done this kind of thing before

Set up in Desktop or HTTPS SSH https://github.com/semmytaju/front-end-web.git

Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

...or create a new repository on the command line

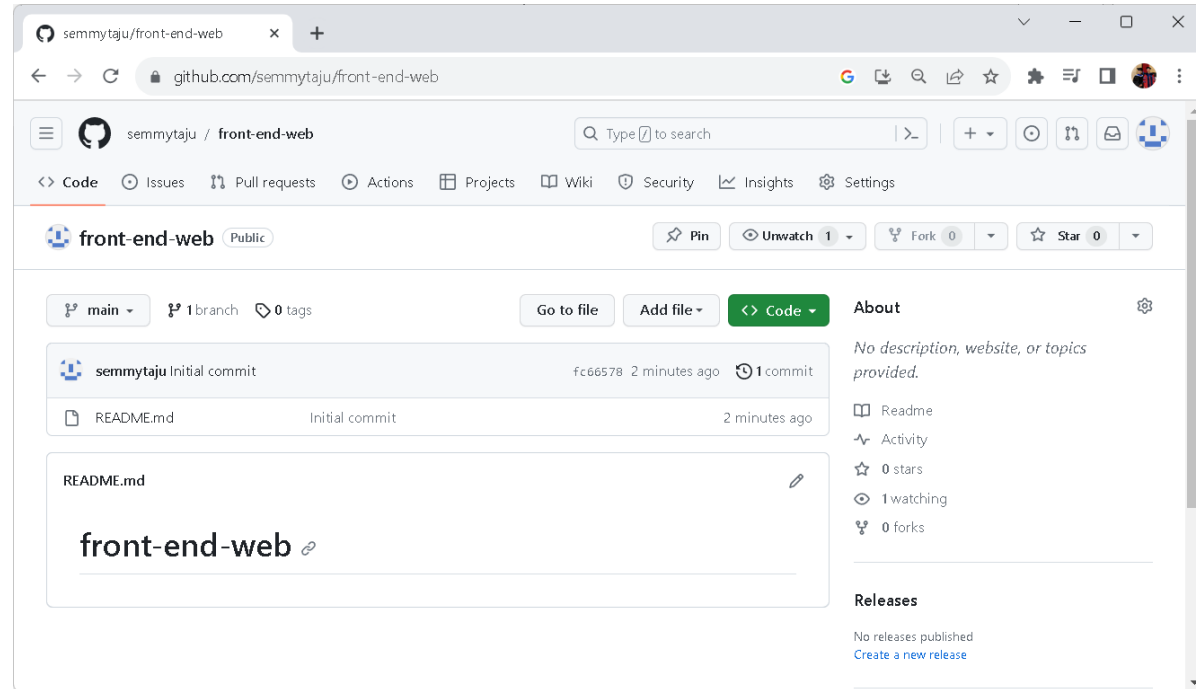
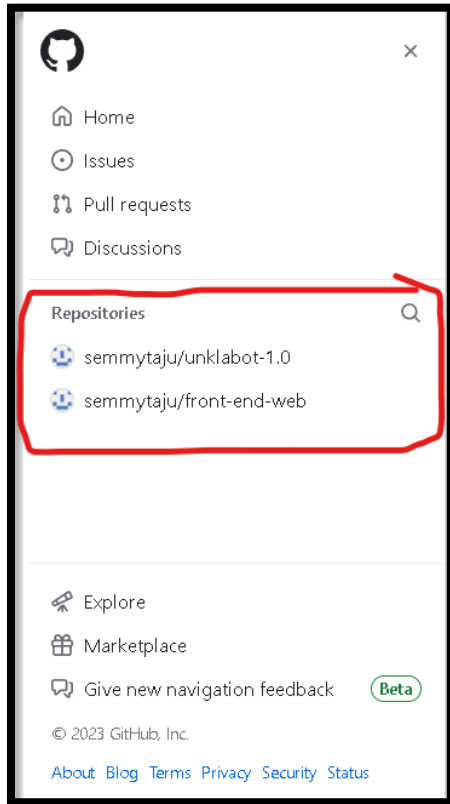
```
echo "# front-end-web" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/semmytaju/front-end-web.git
git push -u origin main
```

...or push an existing repository from the command line

```
git remote add origin https://github.com/semmytaju/front-end-web.git
git branch -M main
git push -u origin main
```

...or import code from another repository

Repository Created



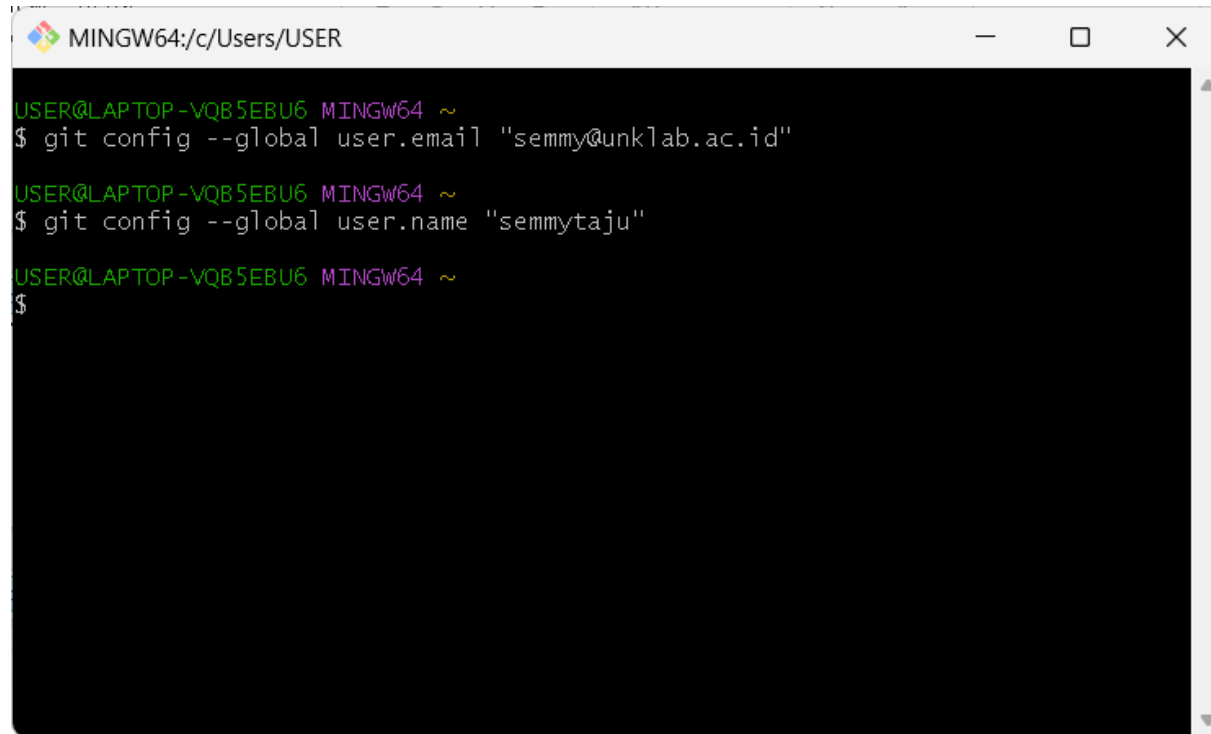
Setting Author Identity !!!

Author identity unknown (Please tell me who you are?). jika baru pertama install git maka kita konfigurasi kan email dan username git di lokal komputer kita untuk footprint atau jejak kita.

Run Following Script:

- `git config --global user.email "semmy@unklab.ac.id"`
- `git config --global user.name "semmytaju"`

Sesuaikan dengan kalian punya **email** dan **username**.

A screenshot of a MINGW64 terminal window. The title bar shows the path 'MINGW64:/c/Users/USER'. The terminal content shows three lines of command execution: the first line shows the prompt 'USER@LAPTOP-VQB5EBU6 MINGW64 ~' followed by '\$ git config --global user.email "semmy@unklab.ac.id"'; the second line shows the same prompt followed by '\$ git config --global user.name "semmytaju"'; and the third line shows the prompt followed by '\$'.

```
MINGW64:/c/Users/USER
USER@LAPTOP-VQB5EBU6 MINGW64 ~
$ git config --global user.email "semmy@unklab.ac.id"
USER@LAPTOP-VQB5EBU6 MINGW64 ~
$ git config --global user.name "semmytaju"
USER@LAPTOP-VQB5EBU6 MINGW64 ~
$
```

Example Directories

Berikut ini adalah beberapa contoh directory yang akan diupdate ke Github directory.

Folder Location: **D:/test/front-end/**

Open Git Bash:

Select/choose local drive:

➤ **cd d:**

Select/choose local folder:

➤ **cd test**

Select/choose local folder:

➤ **cd front-end**

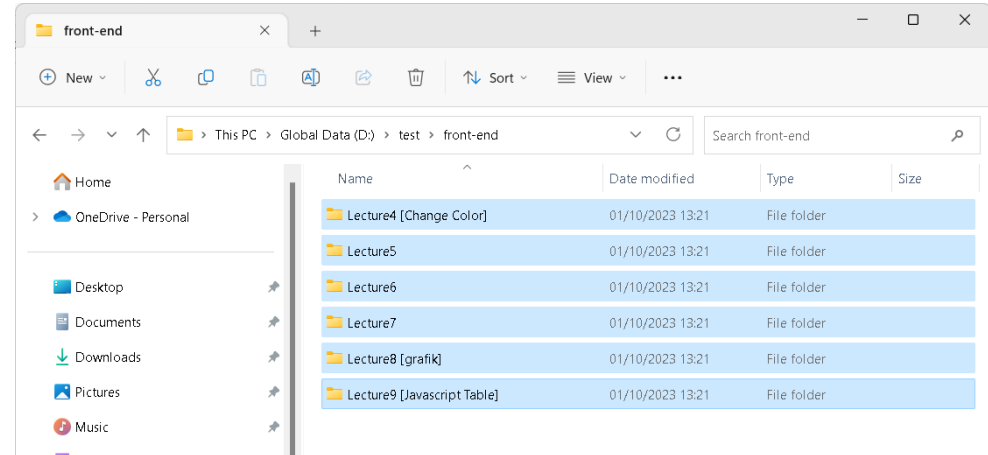
View directory:

➤ **dir**

List directory:

➤ **ls**

“Sesuaikan dengan lokasi dan nama folder kalian.”



```
MINGW64;d/test/front-end

USER@LAPTOP-VQB5EBU6 MINGW64 ~
$ cd d:

USER@LAPTOP-VQB5EBU6 MINGW64 /d
$ cd test

USER@LAPTOP-VQB5EBU6 MINGW64 /d/test
$ cd front-end/

USER@LAPTOP-VQB5EBU6 MINGW64 /d/test/front-end
$ dir
Lecture4\ [Change\ Color]  Lecture6  Lecture8\ [grafik]
Lecture5                  Lecture7  Lecture9\ [Javascript\ Table]

USER@LAPTOP-VQB5EBU6 MINGW64 /d/test/front-end
$ ls
'Lecture4 [Change Color]'/  Lecture6/  'Lecture8 [grafik]'/
Lecture5/                  Lecture7/  'Lecture9 [Javascript Table]'/

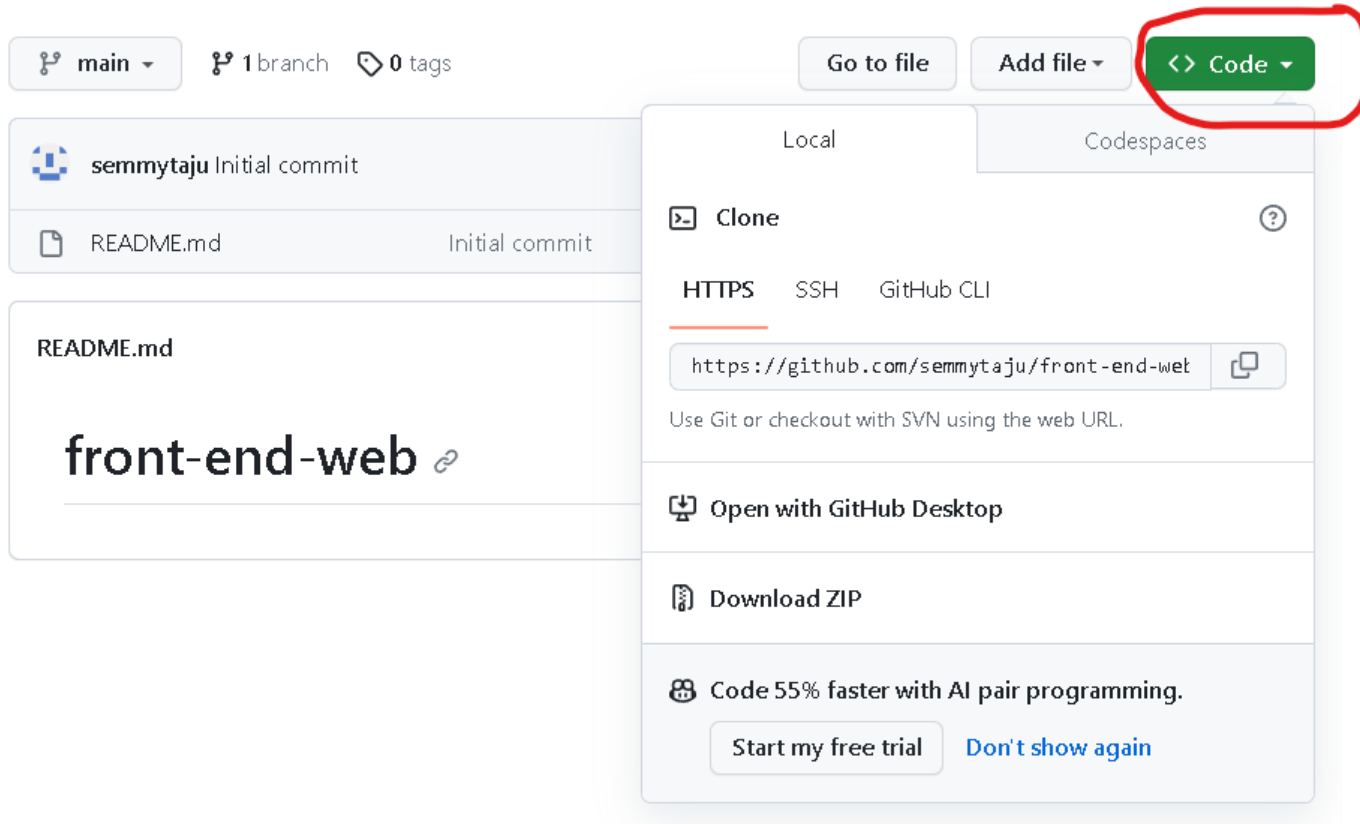
USER@LAPTOP-VQB5EBU6 MINGW64 /d/test/front-end
$
```

Understand Git Commands

- 1) **git init**: Perintah ini digunakan untuk menginisialisasi repository Git baru. Ini membuat direktori kosong menjadi repositori Git yang siap untuk digunakan.
- 2) **git add .**: Perintah ini digunakan untuk menambahkan semua perubahan yang ada dalam direktori kerja Anda ke staging area. Dalam kasus ini, tanda "." mengacu pada semua file dan folder yang ada di direktori saat ini.
- 3) **git commit -m "commit pertama saya"**: Setelah perubahan ditambahkan ke staging area, kita perlu melakukan commit untuk menyimpan perubahan tersebut ke dalam sejarah repositori. Dalam perintah ini, kita memberikan pesan commit sebagai "*commit pertama saya*" yang digunakan untuk memberikan deskripsi singkat tentang perubahan yang dilakukan.
- 4) **git branch -m main**: Perintah ini digunakan untuk mengganti nama branch saat ini ke "main". Sebelumnya, **branch default dalam repositori Git adalah "master"**, tetapi banyak repositori telah beralih ke menggunakan "main" sebagai nama branch default.
- 5) **git remote add origin <https://github.com/semmytaju/front-end-web.git>**: Perintah ini digunakan untuk menambahkan remote repository dengan nama "origin" yang terhubung ke URL yang diberikan. Ini adalah langkah yang diperlukan jika kita ingin menghubungkan repositori lokal dengan repositori di GitHub atau server Git lainnya.
- 6) **git push -u origin main**: Perintah ini digunakan untuk mengirimkan perubahan yang sudah kita commit ke remote repository (GitHub dalam kasus ini). "-u" digunakan untuk mengatur "origin/main" sebagai branch default untuk melakukan push di masa depan. Artinya, kita bisa melakukan "git push" tanpa perlu menyebutkan branch lagi setiap kali.

Type Access To Github

Untuk menghubungkan **repositori lokal** dengan repositori di GitHub atau server Git maka kita perlu tipe akses yang sesuai. Disini saya memilih cara yang paling mudah yaitu dengan URL HTTPS.



A terminal window with a black background and green text. It shows a user navigating through directories and initializing a Git repository. The commands include 'cd d:', 'cd test', 'cd front-end/', 'git init', 'git commit -m "commit pertama saya"', and 'git add .' followed by listing files. To the right of the terminal, there are three large, bold, black text boxes containing instructions in Indonesian: 'Open Git Bash:', 'Menginisialisasi repositori' (with a red arrow icon), 'Menambahkan semua pe...' (with a red arrow icon), 'Menyimpan perubahan:' (with a red arrow icon), and 'Mengganti nama branch' (with a red arrow icon).

```
USER@LAPTOP-VQB5EU6 MINGW64 ~
$ cd d:

USER@LAPTOP-VQB5EU6 MINGW64 /d
$ cd test

USER@LAPTOP-VQB5EU6 MINGW64 /d/test
$ cd front-end/

USER@LAPTOP-VQB5EU6 MINGW64 /d/test/front-end (master)
$ git init
Reinitialized existing Git repository in D:/test/front-end/.git/

USER@LAPTOP-VQB5EU6 MINGW64 /d/test/front-end (master)
$ git add .

USER@LAPTOP-VQB5EU6 MINGW64 /d/test/front-end (master)
$ git commit -m "commit pertama saya"
[master (root-commit) c0c5f5a] commit pertama saya
33 files changed, 742 insertion
create mode 100644 Lecture4 [ch
create mode 100644 Lecture4 [ch
create mode 100644 Lecture5/dem
create mode 100644 Lecture5/dem
create mode 100644 Lecture5/dem
create mode 100644 Lecture5/dem
create mode 100644 Lecture5/dem
create mode 100644 Lecture5/dem
create mode 100644 Lecture5/dem
create mode 100644 Lecture5/dem
create mode 100644 Lecture5/dem
create mode 100644 Lecture6/dem
create mode 100644 Lecture6/dem
create mode 100644 Lecture6/dem
create mode 100644 Lecture6/dem
create mode 100644 Lecture6/dem
create mode 100644 Lecture7/dem
create mode 100644 Lecture7/dem
create mode 100644 Lecture7/dem
create mode 100644 Lecture8 [gr
create mode 100644 Lecture8 [gr
create mode 100644 Lecture8 [gr
create mode 100644 Lecture8 [gr
create mode 100644 Lecture8 [gr
create mode 100644 Lecture8 [gr
create mode 100644 Lecture9 [ja
create mode 100644 Lecture9 [ja
create mode 100644 Lecture9 [ja
```

Open Git Bash:

Menginisialisasi repositori

➤ **git init**

Menambahkan semua pe

➤ **git add .**

Menyimpan perubahan:

➤ **git commit -m "com**

Mengganti nama branch

➤ **git branch M mai**

Open Git Bash:

➤ `git init`

➤ `git add .`

➤ `git commit -m "commit pertama saya"`

➤ `git branch -M main`

➤ git remote add origin <https://github.com/semmytaju/front-end-web.git>

➤ `git push -u origin main`

Get Push (kirim perubahan)

```
MINGW64:/d/test/front-end
USER@LAPTOP-VQB5EBU6 MINGW64 /d/test/front-end (main)
$ git push -u origin main
error: src refspec origin does not match any
error: failed to push some refs to '-u'

USER@LAPTOP-VQB5EBU6 MINGW64 /d/test/front-end (main)
$ git push -u origin main
```

Form login ke
Github akan
muncul secara
otomatis.



Pilih “*Sign In with you browser*”.

Connect to GitHub

GitHub

Sign in

Browser/Device

Token

Sign in with your browser

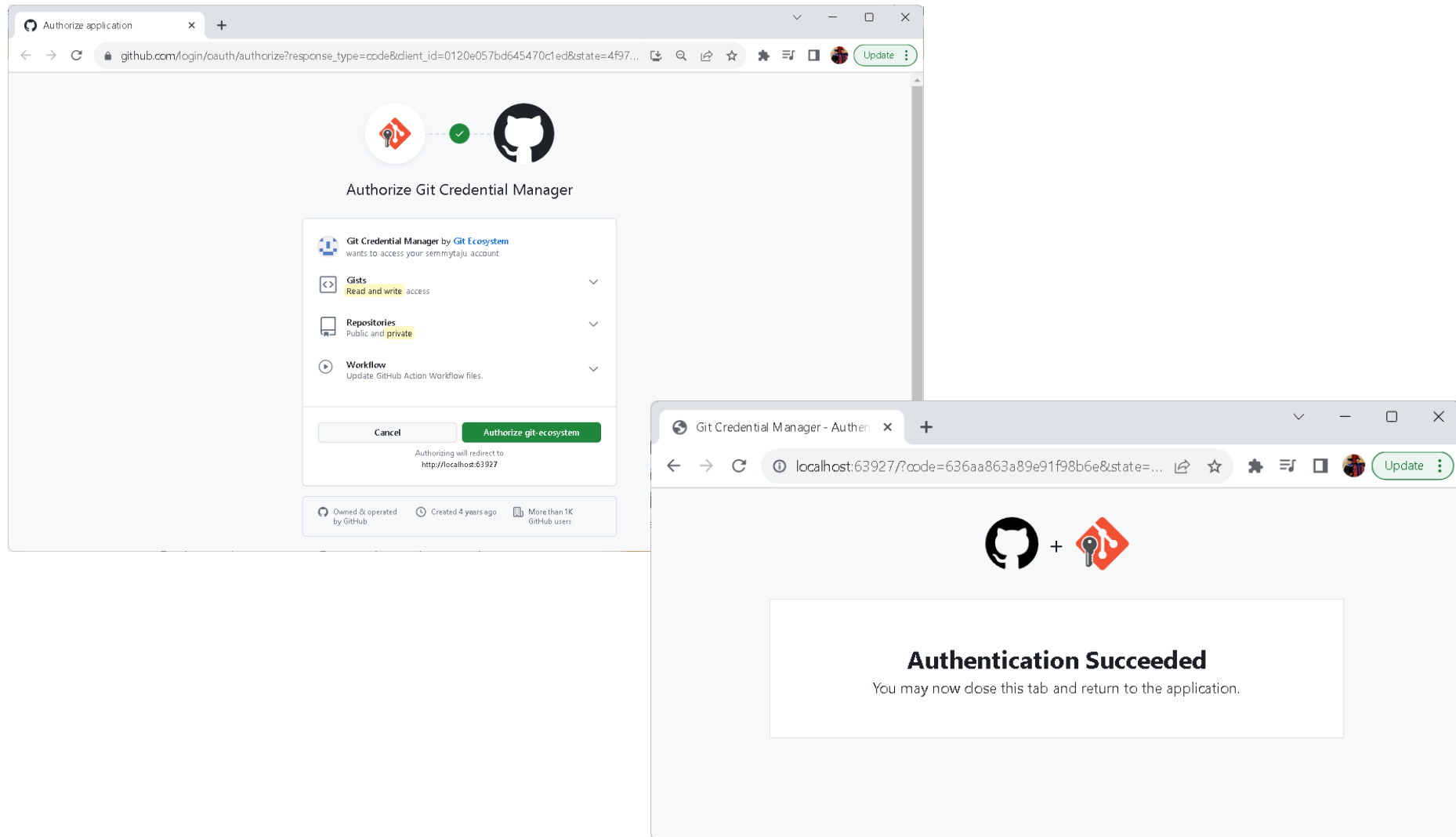
Sign in with a code

Don't have an account?

[Sign Up](#)

Authorize Git Credential Manager

Pilih “Sign In with you browser”.



Get Push Successfully

```
MINGW64:/d/test/front-end
USER@LAPTOP-VQB5EBU6 MINGW64 /d/test/front-end (main)
$ git push --help

USER@LAPTOP-VQB5EBU6 MINGW64 /d/test/front-end (main)
$ git push -u origin main
Enumerating objects: 62, done.
Counting objects: 100% (62/62), done.
Delta compression using up to 2 threads
Compressing objects: 100% (54/54), done.
Writing objects: 100% (62/62), 15.77 KiB | 950.00 KiB/s, done.
Total 62 (delta 4), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (4/4), done.
To https://github.com/semmytaju/front-end-web.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

USER@LAPTOP-VQB5EBU6 MINGW64 /d/test/front-end (main)
$

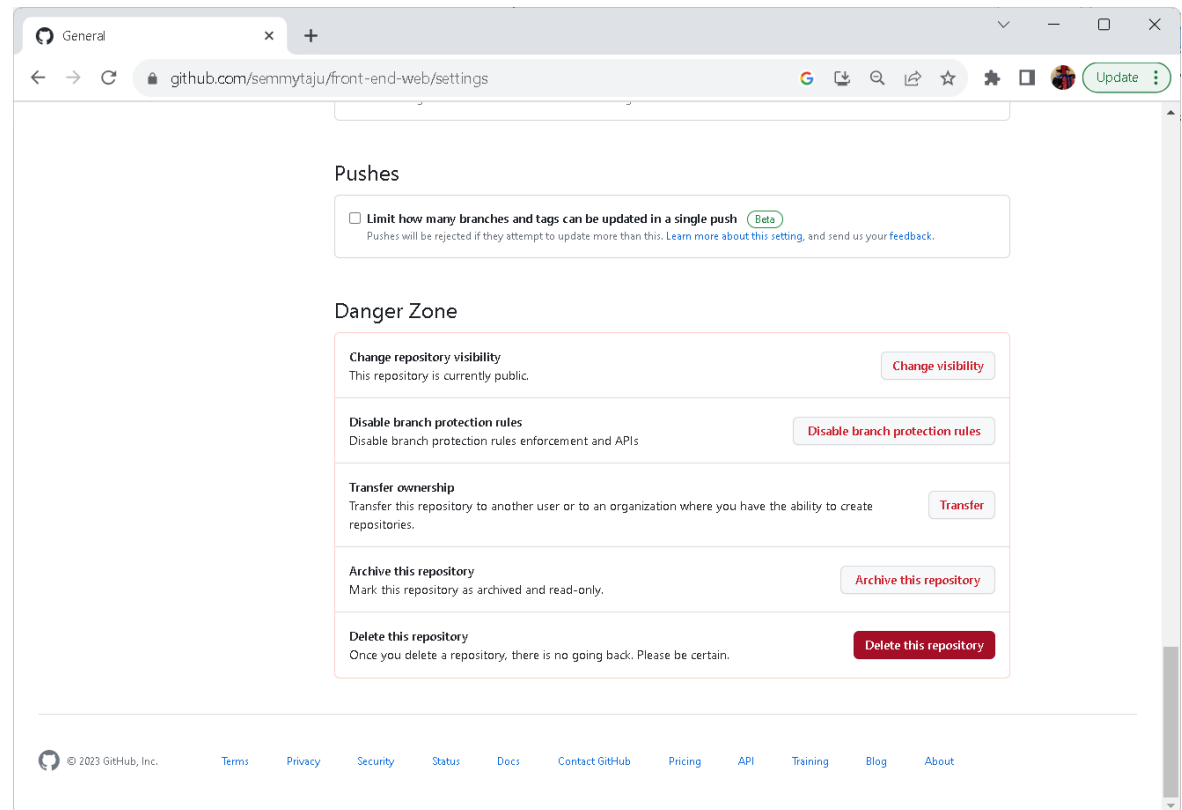
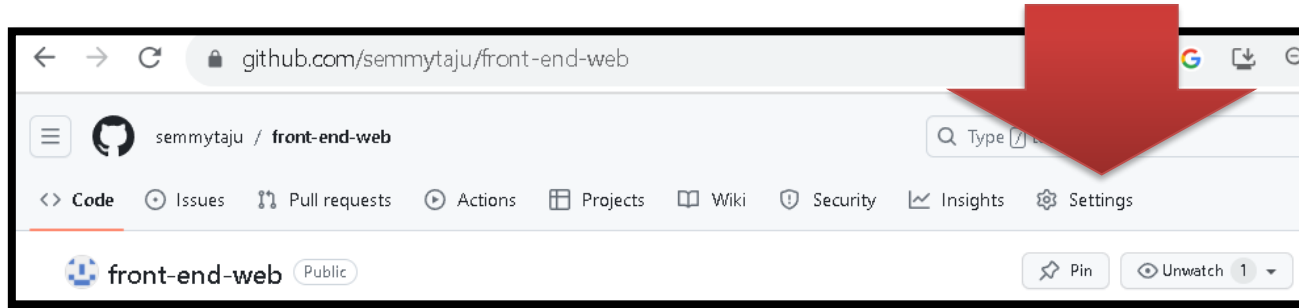
USER@LAPTOP-VQB5EBU6 MINGW64 /d/test/front-end (main)
$

USER@LAPTOP-VQB5EBU6 MINGW64 /d/test/front-end (main)
$ |
```

The screenshot shows the GitHub repository page for 'semmytaju / front-end-web'. The repository is public and has 1 branch (main) and 0 tags. The commit history shows a series of commits, each labeled 'commit pertama saya' and dated '45 minutes ago'. The repository has 0 stars, 1 watching, and 0 forks. The 'About' section is empty. The 'Releases' section shows 'No releases published' with a link to 'Create a new release'. The 'Packages' section shows 'No packages published' with a link to 'Publish your first package'. The 'Languages' section shows a bar chart with HTML at 48.6%, JavaScript at 46.6%, and CSS at 4.8%.

Commit	Author	Date
commit pertama saya	semmytaju	45 minutes ago
Lecture4 [Change Color]/test-demo-j...	commit pertama saya	45 minutes ago
Lecture5	commit pertama saya	45 minutes ago
Lecture6	commit pertama saya	45 minutes ago
Lecture7/demo3	commit pertama saya	45 minutes ago
Lecture8 [grafik]/demo8	commit pertama saya	45 minutes ago
Lecture9 [Javascript Table]/demo9	commit pertama saya	45 minutes ago

Privately Share & Delete Repository





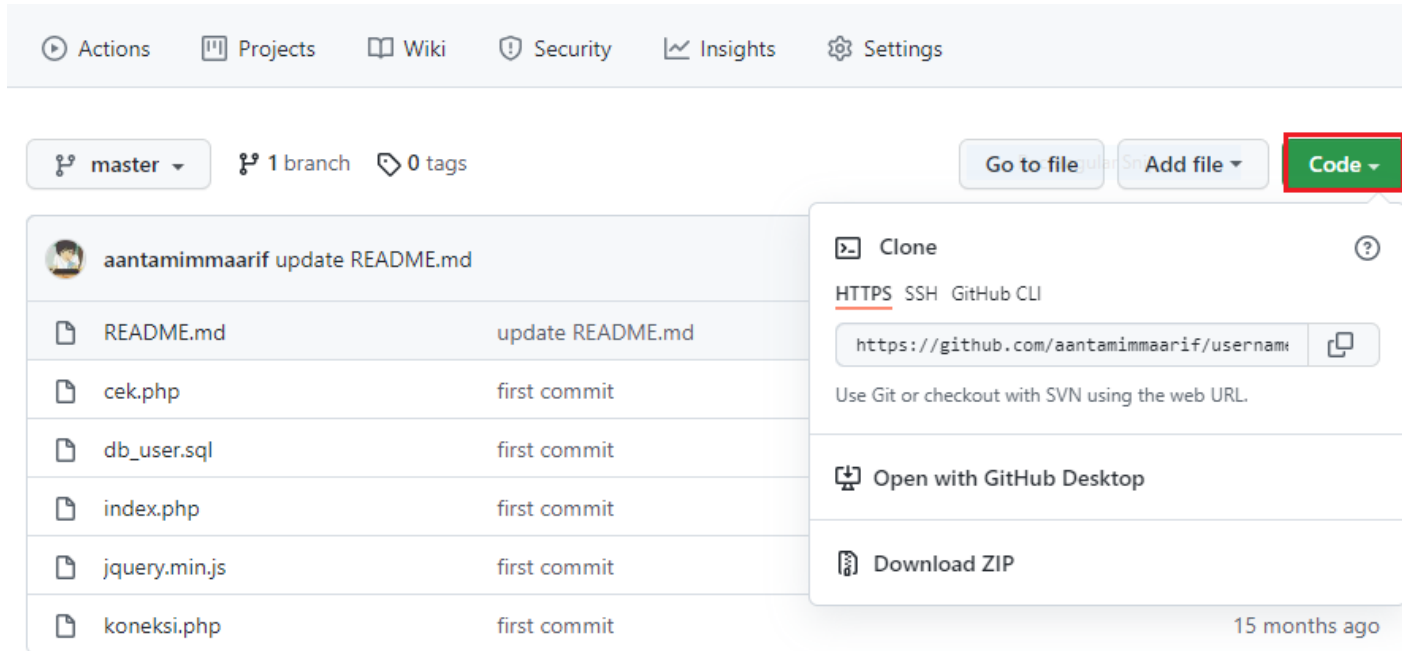
6 – Clone Repository GitHub

(git clone = membuat salinan/copy ke repository local)



Clone Repositori GitHub

Kita akan mencoba melakukan **CLONE** sebuah repositori di GitHub, click tombol berwarna hijau “Code” dan pilih tab **HTTPS** kemudian salin URL yang berakhiran .git tersebut.



The screenshot shows a GitHub repository page for user 'aantamimmaarif'. The repository has 1 branch (master) and 0 tags. The file list includes README.md, cek.php, db_user.sql, index.php, jquery.min.js, and koneksi.php. The 'Code' button is highlighted with a red box, and its dropdown menu is open, showing options for cloning the repository via HTTPS, SSH, or GitHub CLI, opening with GitHub Desktop, or downloading a ZIP file.

Actions Projects Wiki Security Insights Settings

master 1 branch 0 tags

Go to file Add file Code

Clone

HTTPS SSH GitHub CLI

<https://github.com/aantamimmaarif/username>

Use Git or checkout with SVN using the web URL.

Open with GitHub Desktop

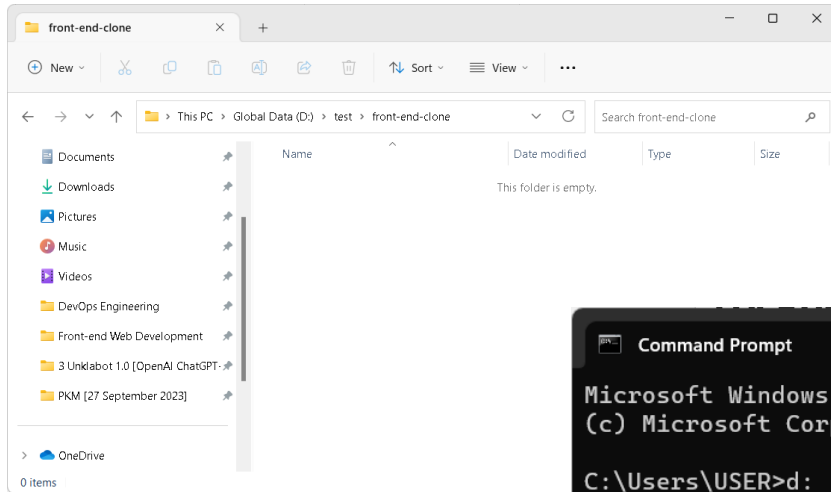
Download ZIP

15 months ago

File	Commit
README.md	update README.md
cek.php	first commit
db_user.sql	first commit
index.php	first commit
jquery.min.js	first commit
koneksi.php	first commit

Create New Local Folder

Location: **D:/test/front-end-clone/**



“Sesuaikan dengan lokasi dan nama folder kalian.”

Open CMD Windows:

Select/choose local drive:

➤ **d:**

Select/choose local folder:

➤ **cd test**

Select/choose local folder:

➤ **cd front-end-clone**

Clone github directory:

➤ **git clone https://github.com/semmytaju/front-end-web.git**

```
Command Prompt
Microsoft Windows [Version 10.0.22621.2283]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER>d:

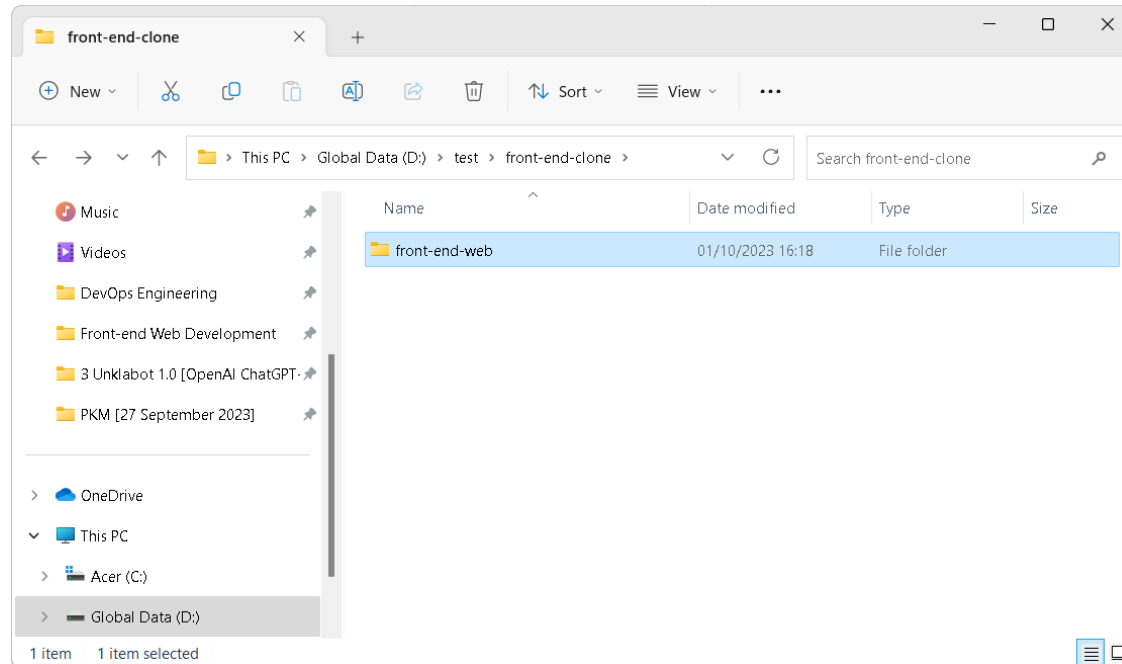
D:\>cd test

D:\test>cd front-end-clone

D:\test\front-end-clone>git clone https://github.com/semmytaju/front-end-web.git
Cloning into 'front-end-web'...
remote: Enumerating objects: 62, done.
remote: Counting objects: 100% (62/62), done.
remote: Compressing objects: 100% (50/50), done.
Receiving objects: 100% (62/62), 15.77 KiB | 181.00 KiB/s, done.
Resolving deltas: 100% (4/4), done.

D:\test\front-end-clone>
```

Repository Github Copied



Clone Repository menggunakan SSH Github:

<https://aantamim.id/git-windows-install/>

Cara Berkontribusi di dalam Project Open Source Github:

<https://www.petanikode.com/github-workflow/>



Cara Install Git & Git Bash di Windows:

<https://youtu.be/ligsdfdR7vI>

Cara Upload Folder & File ke Github:

<https://youtu.be/WRsK4f9fmY8>

A decorative border of green leaves and branches at the top of the page.

7 – Push ke GitHub dari Ubuntu 22.04

A decorative border of green leaves and branches at the bottom of the page.

Install & Setting Identitas Git

Menginstal Git di Ubuntu 22.04 Server:

- `sudo apt install git`
- `git --version`

Setting Identitas Git:

- `git config --global user.name "Nama Anda"`
- `git config --global user.email "alamat@email.com"`

Push ke GitHub:

- `mkdir testdirdevops`
- `git add .`
- `git commit -m "Pesan commit Anda"`
- `git push origin master`

Clone dari GitHub ke Ubuntu 22.04 Server:

- `git clone <URL Repositori>`



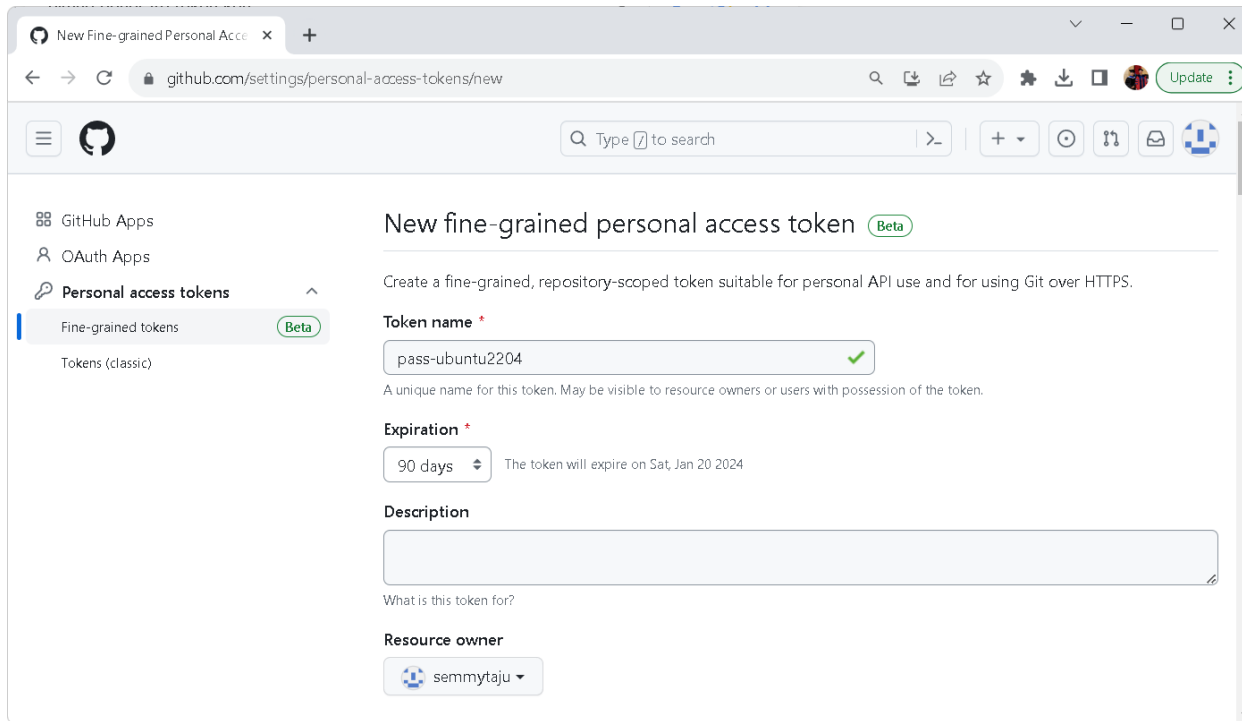
Exercise *for* Students

Class Activities

- 1) Students diharapkan untuk bisa menginstall Git di ubuntu 22.04.
- 2) Students diharapkan bisa membuat new account di Github.
- 3) Students bisa melakukan **PUSH** semua *folder exercises* yang dibuat dari Ubuntu 22.04 ke Github.
- 4) Students bisa melakukan **CLONE** repository di github ke *local folder* di Ubuntu 22.04.

Personal Access Tokens (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a **password for Git over HTTPS**, or can be used to authenticate to the API over Basic Authentication.

A screenshot of the GitHub web interface showing the 'New fine-grained personal access token' page. The browser address bar shows 'github.com/settings/personal-access-tokens/new'. The left sidebar contains a menu with 'GitHub Apps', 'OAuth Apps', 'Personal access tokens' (selected), 'Fine-grained tokens' (marked with a 'Beta' badge), and 'Tokens (classic)'. The main content area is titled 'New fine-grained personal access token' with a 'Beta' badge. Below the title is a description: 'Create a fine-grained, repository-scoped token suitable for personal API use and for using Git over HTTPS.' The form includes a 'Token name' field with the value 'pass-ubuntu2204' and a green checkmark, a note that it's a unique name, an 'Expiration' dropdown set to '90 days' with a note that the token will expire on 'Sat, Jan 20 2024', a 'Description' text area, and a 'Resource owner' dropdown set to 'semmytaju'.

Creating a personal access token:

- 1) In the upper-right corner of any page, click your profile photo, then click Settings.
- 2) In the left sidebar, click Developer settings.
- 3) In the left sidebar, click Personal access tokens.
- 4) Click Generate new token.

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

